

VidhyaSaathy

YOUR COMPANION IN LEARNING

COMPLETE APPLICATION DOCUMENTATION · BREAKTHROUGH EDITION

VidhyaSaathy

"Your school, online, after the bell rings."

The complete, consolidated documentation of the VidhyaSaathy platform — an AI-powered School-to-Home Learning Continuity Platform built for Indian schools. Every mode, feature, workflow and dashboard for **students, teachers, parents and school administrators**, from sign-up and sign-in through the full learning loop: all **100 numbered features (VS-01 to VS-100)**, the AI engine, the India layer, end-to-end flows, architecture, build order and acceptance criteria — nothing invented, nothing important omitted.

BUILT FOR FOUR CONNECTED ROLES

Students · Teachers · Parents · School Admins

101	100	18	5
FEATURES · ONE LOOP	VS FEATURE IDS	DOCUMENTATION PARTS	SOURCE DOCUMENTS

Version 1.0 Final · 19 October 2026 · Classes 9–10 prototype, designed to expand across grades
Vikrant University Junior Hackathon 2026 team workspace · Consolidated from five source documents (see Part 18.4)

Contents

01 **Product Foundation** The core learning loop · the problem · ten principles · roles · boundaries · the moat — 1

02 **Brand, Theme and the Cloud Companion** Mascot & expressions · the official palette · interface rules · logo · companion states — 6

03 **Entry System: Sign-Up, Sign-In, Onboarding** VS-01–08 · onboarding wizard · accounts for every role · RBAC · screen map — 8

04 **The Class Workspace** VS-09–14 · structured class home · tagged notes · segmented recordings · chapter history — 13

05 **Student World: Dashboard and Learning** VS-15–27 · guided self-study sessions · homework · quizzes · recovery path · concept map — 15

06 **The AI Engine** VS-28–59 + Blueprint AI · honesty layer · Socratic mode · gap detector · Syllabus GPS — 21

07 **Teacher World** VS-60–72 · dashboard · heat-map · session builder · auto-grading · AI-drafted sessions — 32

08 **Parent World** VS-73–78 · plain-language dashboard · summaries · ask-this · WhatsApp · privacy rules — 35

09 **School Admin World** VS-79–87 · structure not surveillance · calendar · CSV enrolment · question bank — 38

10 **The India Layer** VS-88–92 · offline + smart sync · data saver · language layers · festival calendar — 40

11 **Motivation, Games and Learning Design** VS-93–100 · streaks with freeze · improvement badges · daily challenge · five games — 41

12 **End-to-End Flows: The Whole System in Motion** A day in Class 9-A minute by minute · the Motion worked example · the loop in practice — 43

13 **Data, Security and Responsible AI** Permission enforcement · data protection · AI safety · evidence integrity — 45

14 **Technical Architecture** Starting stack · 37 data entities · AI request workflow · failure and fallback — 46

15 **Build Order, Phases and Deliberate Cuts** Master Spec phases 1–4 · Blueprint priorities 1–5 · deliberate cuts · one-line truth — 47

16 **Acceptance Criteria and QA Checklists** Account & permission · learning workflow · AI reliability · UX · validation tests — 49

17 **Feature Registry and Cross-References** VS-01–100 index · role × feature matrix · Blueprint modules · open decisions · chains — 50

18 **Glossary, Team Workspace and Closing** Product vocabulary · team protocol · final definition · sources · fidelity note — 57

HOW TO NAVIGATE THIS DOCUMENT

Feature IDs — VS-01 to VS-100
Every feature carries its permanent ID from the Master Working Spec, so the team can discuss "How does **VS-42** work?" without ambiguity. The full index is in Part 17.

Status tags
CORE founder-confirmed · PROPOSED specified, awaiting approval · OPEN explicitly undecided · P1–P4 build phase · ★ source emphasis (★ highlighted, ★★ flagship, ★★★ highest).

Cross-references
Features reference each other as VS-30, sections as (Section 7.2), and Blueprint PRD modules as (S5 §53). Six key workflow chains are mapped in Section 17.5.

Checklists
Every part ends with an implementation checklist. The master acceptance checklists are consolidated in Part 16. Undecided items stay
OPEN
exactly as in the sources.

PART 1 · FOUNDATION

Product Foundation

What VidhyaSaathy is, the single loop every feature orbits, the problem it solves, the principles that govern every design decision, and the boundary between what the product is and is not.

1.1 What VidhyaSaathy Is

VidhyaSaathy is a **school-connected learning platform** — an online copy of each school class. Whatever happens in the offline classroom (notes, homework, tests, lesson recordings) also appears online in that class's workspace, so the school keeps helping the student at home. It is formally defined in the Blueprint as an **AI-powered School-to-Home Learning Continuity Platform**.

A school already provides teachers, textbooks, explanations, notes, assignments, tests and guidance — but once students return home, they must independently decide what to study, which resources to trust, how to understand difficult topics and how to recover when they fall behind. Students switch between YouTube videos, PDFs, friends' notes, websites and separate AI tools without knowing which explanation matches their school's teaching. VidhyaSaathy connects these disconnected activities into one system:

- **One workspace per class** (for example Class 9-A). Students, teachers and parents are all connected to it.
- **The teacher leads the class**: they publish notes, homework, recordings and guided self-study sessions.
- **Self-study sessions are fixed, structured units.** The school makes them available; the teacher decides the day they open.
- **An AI helper** answers questions in the context of the subject, class and the teacher's own material, and builds a study plan that points to the right school note, recording or video.
- **Parents see what the child is studying and what is completed** — not every private conversation.

> *The one-line thesis of the whole product: "Why should the school's support end when the bell rings?" — the school speaks again after the bell rings.*

Students should be able to answer four questions at any moment: **What should I study today? What do I still not understand? How do I practise it? What should I do next?**

1.2 The Core Learning Loop — the App's Soul

The entire app runs a single loop; all ~101 features orbit around it:

TEACHER TEACHES IN CLASS

↓

NOTES / SESSION ARE PUBLISHED
VS-11 · VS-62 · VS-63

↓

STUDENT STUDIES AT HOME
VS-15 · VS-16

↓

STUDENT PRACTISES
VS-17 · VS-18

↓

EVIDENCE IS COLLECTED
VS-27 · VS-39–VS-42

↓

AI SUGGESTS THE NEXT STEP
VS-28–VS-31 · VS-58

↓

TEACHER SEES CLASS-LEVEL GAPS
VS-60 · VS-61 · VS-37

↓

NEXT CLASS IS BETTER → LOOP REPEATS

The extended version from the Blueprint adds the parent: the teacher sees class-level patterns → the parent receives a meaningful progress update (**VS-69** , **VS-74**) → the student continues learning.

DESIGN PRINCIPLE

The teacher's effort is applied **once**; the system reuses it again and again; the AI adds school context; evidence decides the next step. The product becomes more useful every time the loop repeats, because it uses reliable evidence from previous sessions.

1.3 The Problem Being Solved

The problem is not that learning material is missing. The problem is that there is **too much of it, and no path that says what to do next**.

Problem	What it looks like for a student	Source
Too many sources	YouTube, PDFs, friends' notes, AI tools, coaching notes — different versions, no way to know which matches the school syllabus.	Founder observation
No clear direction	"Okay, I have to study Physics tonight. What do I even do?"	Founder observation
No check of understanding	An hour of study does not show whether the concept was understood.	Founder observation
Missed lessons	A missed class or weak prerequisite creates gaps that grow later.	Reasoned, needs testing
Coaching gap	Students with coaching get a second structure; students without it manage alone.	Hypothesis, needs research
Repetitive writing	Much class time goes to copying notes from the board, for students and teachers.	Founder observation
Parents in the dark	Parents see marks, not what the child studies or where they are stuck.	Hypothesis, needs research
Distraction	Students open many slides at once, switch apps, or skip the test.	Founder observation

These are the founder's observations and hypotheses, not yet backed by surveys or published research. Validation is tracked in the team workspace (Part 18) and the open-decisions register (Section 17.4).

1.4 Where the Idea Came From

The founder was a strong student, scoring above 90% in earlier classes. In Class 9, studying became chaotic. There was no coaching, so when a topic like Biology or Physics was hard, the founder searched YouTube, asked friends for notes, and used AI tools. Friends' "perfect notes" were all different, and the founder had to decide what to trust. The lesson carried into the product: the problem was not that nothing was available — it was that nothing came together into one coherent path. This is one student's experience, not proof that every student faces the same problem, which is why the sources mark market claims as hypotheses to validate. The first target is schools in Indian Tier-2 and Tier-3 cities, as an initial interest rather than a validated market.

1.5 The Ten Product Principles

From the Blueprint (S5 §2) — these govern every design decision in this document:

1 · School remains central

The school's syllabus, teachers and approved learning resources form the foundation of the experience.

2 · AI supports teachers, never replaces them

Teachers remain responsible for curriculum decisions, important assessments and interventions.

3 · Personalisation is based on evidence

The app responds to demonstrated difficulties; it never permanently labels students as weak or intelligent.

4 · Understanding beats activity counts

Completing ten activities is not equivalent to mastering a concept.

5 · Parents receive insights, not just marks

They should understand what the child has learned and how they can help.

6 · Works for real Indian contexts

Shared devices, different languages, variable connectivity, limited teacher time.

7 · Calm and encouraging

No shame-based rankings, manipulative streaks or excessive notifications.

8 · AI output is grounded and reviewable

Traceable to school material or another identified source.

9 · Privacy is a product requirement

A child's private questions do not automatically become a parent-facing transcript.

10 · Every feature connects to the loop

Identify a learning need, take an action, check whether the action worked.

1.6 Users and Roles

Role	How they join	What they do & see
Student	Roll/admission number + class, section + activation code + parent-phone OTP (VS-04)	Reads notes, does self-study, homework and quizzes, asks the AI helper. Sees own class workspace, own study plan, own progress only.
Teacher	Staff ID + temporary password from admin (VS-03)	Publishes notes, creates and times sessions, sets homework and quizzes, shares recordings. Sees assigned classes, completion, concepts many students miss.
Class teacher	A teacher with extra rights for one class (VS-72)	Manages the class workspace, broadcasts class updates. Whole-class view plus class-space management.
Parent	Linked to a verified child via OTP (VS-05)	Follows progress and tasks. Own child only — never private AI chats.
School admin	School owner account (VS-02)	Creates classes, sections, teacher assignments and enrolments. School-wide structure — never student rankings.
Platform admin	Platform operations staff (distinct from school admin)	Security, service health, authorised support. Only access required for legitimate operations.

1.7 Product Boundaries — What It Is / Is Not

VidhyaSaathy IS

A school-connected learning ecosystem · a bridge between classroom and home study · a structure for teacher-guided self-learning · a way to link resources to assessments · a tool to show where support is needed · a way to keep families and schools connected.

VidhyaSaathy IS NOT

A generic AI chatbot · a homework-answer generator · a replacement for teachers · a repository of PDFs and videos · a video-conferencing or attendance app · an immediate elimination of private coaching.

1.8 Feature IDs, Statuses and the Long-Term Moat

Feature IDs. The Master Working Spec (S4) assigns every feature a permanent ID from **VS-01** to **VS-100** so the team can discuss "How is VS-42?" without ambiguity. The spec describes the catalogue as 101 features (the numbering plus the core loop). This document preserves those IDs exactly and adds Blueprint-only modules with their PRD citations (Section 17.3).

Statuses. Source documents mark commitment levels: CONFIRMED items are part of the idea as the founder described it; PROPOSED items are specifications to discuss; OPEN items are explicitly undecided. This document shows them as CORE, PROPOSED and OPEN, plus build-phase tags P1–P4 from the build order (Part 15).

Long-term moat (S5 §79). No individual feature guarantees a moat. The defensible advantage may emerge from the combination of: (1) school-specific content and curriculum mapping, (2) teacher workflows and adoption, (3) longitudinal, evidence-backed concept histories, (4) misconception patterns connected to actual student responses, (5) intervention history and follow-up outcomes, (6) low-friction school onboarding, (7) parent-teacher-student continuity, and (8) reliable performance on ordinary devices and constrained connections. These are hypotheses to validate through real use — not advantages that can be claimed before implementation.

1.9 Foundation Checklist

- [ ] Team can state the core loop from memory (it drives every screen)
- [ ] Every designed screen maps to at least one loop step
- [ ] No feature labels students (weak / average / smart) anywhere in the UI
- [ ] AI features can name their school-context fuel (notes, quiz results, curriculum position, behaviour log) — otherwise they fail the moat test
- [ ] Claims about markets and outcomes stay marked as hypotheses until validated (Section 16.5)
- [ ] Product boundaries table (1.7) respected in all pitches and demos

PART 2 · BRAND

Brand, Theme and the Cloud Companion

The selected direction, confirmed by the brand board (S3): a friendly cloud mascot hugging a book — soft, modern, approachable. This is the visual language of every screen in the app.

2.1 Brand Identity and Mascot

The mascot is **a cute blue cloud hugging a book and reading**, sitting on stacks of books in the poster art, under the message *"Small steps make big progress."* The brand must feel:

- **Trustworthy** enough for schools and parents.
- **Friendly** enough for students to enjoy using daily.
- **Modern** enough to feel like a new AI learning product.
- **Calm** rather than visually noisy.
- **Distinctive** without copying Duolingo, Chikoo or another existing brand.

Tagline options discussed in the sources (one must be chosen — Section 17.4): "Learn • Grow • Together." (shown on the brand board) · "Grow at your own pace." · "Your school, online, after the bell rings." (the product one-liner).

The mascot has **six named expressions** on the brand board, mapped to product moments:

1 "Let's learn!"
Session start / learning begins.

2 "You got this!"
Encouragement before a difficult question.

3 "Hmm..."
AI thinking / working on an answer.

4 "Keep going!"
Mid-session persistence.

5 "Great job!"
Milestone / quiz success.

6 "Time to rest"
End of day, healthy usage boundary.

The phone UI in the brand board greets the student with "Good morning, Learner!" and shows the three core modules — My Courses, Study Sessions, Practice & Quizzes — matching the dashboard structure in Part 5.

2.2 Colour Palette

Five brand colours — the current direction, not a requirement to use every colour equally:

Sky Blue
#7FB3FF
Primary actions, important learning states, links.

Soft Peach
#FFC8A3
Supportive highlights, celebrations.

Warm White
#FFF9F5
Comfortable reading surfaces.

Deep Navy
#1E3A8A
All readable text.

Light Blue
#E6F0FF
Cards, soft fills, hover states.

The heat-map feature (**VS-61**) reuses palette semantics: green = secure, peach = mixed, red = stuck.

2.3 Interface Rules

≈80% clean, functional interface **≈15%** soft colour **≈5%** playful mascot & illustration

- Use **blue** for primary actions and important learning states; **peach** for supportive highlights and celebrations; **warm white** for comfortable reading; **deep navy** for text.
- Avoid purple/neon-heavy styling, overly cartoonish interfaces and excessive gradients.
- Teacher dashboards may be more information-dense than student screens while preserving the same visual identity.
- Use accessible contrast, readable typography, clear focus states and responsive layouts.

2.4 Logo Requirements

The final logo must have: (1) a recognisable cloud mascot or cloud-based mark — the "V" cloud mark on the brand board is the working direction; (2) a wordmark reading **"VidhyaSaathy"**; (3) a standalone icon for app favicons and small screens; (4) a monochrome version; (5) a version that works on light and dark backgrounds; (6) a simplified version that remains legible at small sizes. The exact final artwork, font and tagline remain to be approved (Open — Section 17.4). The team mockup is titled ClassOS and the product name itself is still OPEN — the idea matters more than the name.

2.5 Cloud Companion Behaviour States

The cloud is a friendly guide throughout the student experience — **not an intrusive chatbot that constantly interrupts**:

State	Behaviour
Idle	Subtle floating animation.
Greeting	Brief welcoming animation when the student opens the learning space.
Thinking	Restrained animation while the AI processes a question.
Encouragement	Small positive reaction after meaningful progress.
Confusion support	Offers help when the student requests it or gets stuck.
Celebration	Brief animation after completing a meaningful milestone.
Resting	Remains unobtrusive while the student reads or solves.
Error	Communicates that something failed without blaming the student.

Reduced motion	Static or minimal animation when requested by device settings.
Consistency	The mascot's appearance stays recognisable everywhere.

HARD RULE

The cloud must not distract students during lessons, flash constantly, or use emotional pressure to keep them online. Voice interaction, emotion detection and advanced animated character behaviour are NOT assumed in the first release — and emotion detection via camera/microphone is deliberately cut forever (Part 15.3).

2.6 Brand Checklist

- [ ] Palette tokens implemented (#7FB3FF / #FFC8A3 / #FFF9F5 / #1E3A8A / #E6F0FF) and 80/15/5 balance respected
- [ ] All six mascot expressions available to the UI state system
- [ ] Mascot never animates distractingly during question-solving
- [ ] Reduced-motion setting honoured globally
- [ ] Logo set (wordmark, icon, mono, dark-bg, small-size) produced and approved
- [ ] One tagline chosen and applied consistently

PART 3 · ENTRY

Entry System: Sign-Up, Sign-In, Onboarding

How every user — school, admin, teacher, student, parent — gets into the system. The school's existing structure (classes, roll numbers, staff IDs) is the app's backbone: the app is a mirror of the school, not a new system.

VS-01 – VS-08

3.1 Public Landing Page

The landing page explains what VidhyaSaathy does before asking anyone to create an account. Its sections (S5 §7): (1) a hero explaining the school-to-home learning promise; (2) student, teacher, parent and school use cases; (3) a visual explanation of the learning loop; (4) a preview of guided learning sessions; (5) a preview of AI-assisted learning and teacher insights; (6) how school material supports AI responses; (7) privacy and child-safety information; (8) school onboarding or contact option; (9) sign-in and account-access links.

GUARDRAIL

The landing page must not claim proven learning improvements, existing school partnerships, or advanced capabilities that have not been validated.

VS-01 **School Onboarding Wizard** CORE P1

The complete setup that brings a school onto the platform — an **8-step wizard** run by the school admin:

1. **Admin registers the school** on VidhyaSaathy.
2. **School profile** — name, board (CBSE / ICSE / State), city.
3. **Academic year** is set.
4. **Classes and sections** are created (e.g. 9-A and 9-B).
5. **Subjects** are added.
6. **Teachers** are assigned to class + subject combinations.
7. **Student list uploaded via CSV** — name, roll number, class, parent phone number.
8. **Activation codes generated** for every student; the school distributes the codes; parents verify via OTP; accounts go live.

Why this way: the school's existing structure is the backbone. App-internal verification of the representative and staged rollout can come later; the first prototype may use a preconfigured demo school. **Cross-ref:** extended by **VS-08** (board mapping at step 2), **VS-83** (enrolment detail), **VS-84** (parent linking), Section 9.5 (CSV validation).

VS-02 School Admin Account & Login

CORE P1

The admin is the school's **owner account**, created during registration. Login with staff ID/email + password; two-step verification optional. The admin sees the full school structure and **NO student marks** — "structure, not surveillance." Admin capabilities are documented in Part 9 (**VS-79** to **VS-87**).

The Blueprint's school registration adds: an authorised representative starts registration → the platform verifies the representative → the school profile is created → admins set up classes → teachers are invited → students are created/invited → parents are linked → subjects configured → first material published (S5 §8).

VS-03 Teacher Account & Login

CORE P1

1. The admin assigns the teacher (subject + class).
2. The teacher receives a staff ID + temporary password.
3. On first login the teacher sets a PIN/password and completes the profile.
4. The dashboard shows ONLY that teacher's assigned classes.
5. Class-teacher role = extra rights for one class (updates, class-space management — **VS-72**).

Creation flow (S5 §10): admin invites or creates the account → teacher verifies invitation → signs in → completes profile → sees assigned classes and subjects → confirms teaching assignments → can upload/create material for authorised classes → sees a short setup guide for the first guided session. Teachers must never automatically see unrelated classes or student records outside their responsibilities.

VS-04 Student Verification & Login — *critical security flow*

CORE · P1 · ★

The principle: students are IDENTIFIED by roll number, but roll number alone is NEVER the login. Roll numbers are public — the whole class knows them. This is children's data; this layer cannot be skipped.

1. Student opens the app → **"Select your school."**
2. Enters roll/admission number.
3. System finds the record but **does not let the user in yet**.
4. System asks for the school-issued **activation code** (one-time use).
5. An **OTP is sent to the parent's registered phone** number.
6. **Both must match.**
7. Student sets a personal **4-digit PIN**.
8. On every later login: school + roll number + PIN only.

Fallback: if the parent's OTP cannot arrive (feature-phone offline, etc.), the admin manually approves the student. **Why it matters:** a student should never see records merely because someone knows a roll number — and registration must not allow an arbitrary user to join a private school class just by knowing its name (S5 §9). Which exact verification combination is default remains OPEN (Section 17.4).

VS-05 Parent Account Linking

CORE · P1

1. The student CSV already carries the parent's phone number.
2. When the student verifies (**VS-04**), the OTP goes to exactly that number.
3. The parent receives an automatic message: *"Your child has joined VidhyaSaathy. Download the app and log in with this number."*
4. The parent creates their account with OTP verification.
5. The parent sees data for ONLY their verified children.
6. Two children? Both are linked; one account, quick switch.

Hard rule: a parent must not be able to connect to a child simply by entering a name or searching a public directory. Guardian cases (uncle, grandfather, etc. — common in India) are handled by **VS-84**.

VS-06 PIN-Based Shared Device Profiles

PROPOSED · P3

Multiple profiles on ONE phone — because in India one phone often serves the whole family. On opening the app, a **"Who are you?"** screen lists the profiles. Each profile has its own PIN; switching takes two taps. If the phone is taken away mid-session, progress is saved — "continue from where you left off."

Cross-ref: connects to **VS-88** (offline sync must respect per-profile data) and the shared-device sign-out reminders in Section 3.10.

VS-07 Role-Based Access Control (RBAC)

CORE P1

One platform, four doors — each role sees only its own work:

Role	Sees / can do
Student	Own class workspace, own study plan, own progress.
Teacher	Only assigned classes; subject teachers only their subject workspace.
Parent	Only their verified child's permitted data.
Admin	School structure only — never student ranking.

Principle: privacy's first door is BLOCKED IN THE DESIGN, not patched later. Enforcement must be server-side (Section 3.10 rule 10; Part 13.1). Changing a client-side role label must never grant additional access.

VS-08 Board-Wise Curriculum Mapping

CORE P1

During onboarding the school selects its board (CBSE / ICSE / State). The platform provides that board's chapter-concept structure ready-made (NCERT listings are public data), so **teachers never build structure from zero**. "Motion" exists in both CBSE and UP Board syllabi, but at different depths. This mapping is the product's lifeblood — it makes tagging (VS-11), gap detection (VS-30) and syllabus GPS (VS-58) possible.

3.10 Sign-In and Session Management Rules

Required behaviours for ALL roles (S5 §12):

1. Email, username or school-issued identifier may be used, according to the chosen authentication configuration.
2. Password-based authentication where appropriate (plus PINs per VS-04 / VS-06 for students).
3. Secure account recovery must be available.
4. Users are routed to their authorised dashboard after authentication.
5. Student accounts must not be able to open teacher or admin routes by changing a URL.
6. Expired sessions return users to a secure sign-in state.
7. Sign-out must end the relevant session.
8. Error messages explain the issue WITHOUT revealing sensitive account information.
9. Shared-device users are reminded to sign out when appropriate.
10. Authentication and permissions are enforced on the SERVER, not only hidden in the interface.

> **CRITICAL**
> The role selected on a login screen is never treated as proof that the user has that role.

3.11 First-Time Onboarding for Every Role

Onboarding is short and useful — it ends with the user's **first real action**.

Student

Welcome to the school learning space (cloud companion greeting) · confirm class and section · select preferred language if offered (**VS-90**) · show today's assigned learning activity immediately · explain how to ask a question or request a different explanation · let the student start learning immediately.

Teacher

Show assigned classes and subjects · explain how to publish a resource · how to create a guided self-study session · how to review student responses · offer a sample class and sample lesson in demo mode.

Parent

Confirm the linked student · explain progress, assignments and notifications · explain the difference between learning progress and private student conversations (**VS-78**) · show the first useful action the parent can take.

3.12 Entry-System Screen Connection Map

LANDING PAGE → SIGN IN → ROLE-AWARE ROUTING → CORRECT DASHBOARD

↓ school path

SCHOOL REGISTRATION → WIZARD

VS-01 · 8 steps

↓

ADMIN DASHBOARD

invite teacher VS-03 → teacher verifies → sets PIN → dashboard

↓

UPLOAD STUDENT CSV → ACTIVATION CODES GENERATED

VS-83

↓

STUDENT: school → roll no → code → parent OTP → set PIN

VS-04 → Student Dashboard (Part 5)

↓

PARENT: auto-invite SMS → OTP

VS-05 → Parent Dashboard (Part 8)

Shared device: the "Who are you?" profile picker (**VS-06**) → profile PIN → role dashboard. Progress resumes where left off.

3.13 Entry-System Checklist

- [ ] Wizard VS-01 runs end-to-end: register → profile → year → classes → subjects → teachers → CSV → activation codes → OTP → live
- [ ] Roll-number-only login is IMPOSSIBLE (activation code + OTP required)
- [ ] Admin manual-approval fallback works when OTP fails
- [ ] Parent sees only verified children; multi-child switch works
- [ ] Shared-device profiles each have independent PINs

- [ ] URL manipulation cannot reach another role's routes (tested in 16.1)
- [ ] Board selected at onboarding pre-loads chapter-concept structure
- [ ] Onboarding for all three roles ends in a real first action
- [ ] Error messages never leak account existence or details

PART 4 · WORKSPACE

The Class Workspace

Every class has a persistent digital home — a mirror of the real classroom where notes, sessions, recordings, homework and quiz results live side by side, organised by curriculum, not chaos.

VS-09 – VS-14

VS-09 Structured Class Workspace

CORE P1

Every class has a persistent digital home — a mirror of the real classroom:

CLASS 9-A → PHYSICS → MOTION (chapter) → DISTANCE & DISPLACEMENT (concept)

↓ everything hangs here

NOTES
VS-11
· **SESSION 3**
VS-16
· **RECORDING (6-min segment)**
VS-12
· **HOMEWORK**
VS-17
· **QUIZ RESULTS**
VS-18

Payoff: "Where do I find the notes?" always has the same answer: your class → subject → chapter → right there. A student in March can still open a February chapter. Not a group chat, not a file dump — an organised learning environment. The intended navigation path from the vision document: School → Class 9-A → Physics → Motion → Lesson → Practice → Assessment → Next learning step.

The workspace organises subjects, chapters, concepts, lessons, notes, recordings, selected external videos, assignments, practice questions, quizzes, study sessions and progress. External resources (a chosen YouTube video) can be attached to a lesson. The platform organises useful resources around the curriculum — it does not try to replace every content platform.

VS-10 Class Space — Same-Class Visibility

CORE

Students and teachers of the same class can see one another — names and profiles — with **no chat spam**. The class teacher shares class updates here: *"Tomorrow's Physics test has been rescheduled."* The class teacher manages the space (VS-72); subject teachers work in their subject workspaces only.

VS-11 Concept-Tagged Teacher Notes

CORE P1

1. Teacher uploads a PDF/PPT.
2. At upload time, tagging takes a few taps: Class → Subject → Chapter → Concept.
3. Publish → the class gets a notification.
4. Notes are available from day one of the chapter.

The real feature is the tagging, not the upload: when Student B is studying "Acceleration" in March, the teacher's definition is two taps away — no waiting for a friend's "perfect notes." Tagging is what later powers **VS-28** (context answers), **VS-30** (gap detection) and **VS-61** (heat-maps).

VS-12 Segmented Classroom Recordings

PROPOSED

1. The lesson is recorded (with school permission — a school-level setting, **VS-85**).
2. The teacher uploads the recording.
3. It is cut into concept segments — manual or AI-assisted (e.g. "Acceleration: 12:40–19:30").
4. Each segment is tagged to its concept.

Payoff: Student C needs "Acceleration" and gets *"Acceleration — 7 minutes"*, not a 45-minute monolith. The recording's value is in segmentation, not in its existence.

VS-13 Trimmed, Curated YouTube Links

PROPOSED

Teacher pastes a link → sets start/end times (e.g. *"watch 8:20–15:40 for this concept"*) → tags the concept → publishes. **Principle:** YouTube is not the enemy — discovery is the problem. What the teacher chooses is correct for the syllabus. Each video resource shows a title and an explanation of why it is relevant. Watching is never counted as mastery (Section 5.16).

VS-14 Persistent Chapter History

CORE

Whatever was taught is **never deleted** — a record of the whole year. After half-yearly exams: "Forgot acceleration?" — reopen it in March: same notes, same session, same quiz. **Payoff:** the school's entire year becomes rewindable — the revision-material job that coaching institutes charge for, included.

4.7 School Content Library and Resource Model

Organisation hierarchy (S5 §16): School → Academic year → Class → Subject → Chapter → Concept → Lesson / resource / activity. A resource record can include: title and description; subject and chapter; intended class and section; resource type; teacher or source; upload date and revision information; publication status; approval status where required; access permissions; related concepts; and source references.

The same resource may be reused across authorised classes rather than uploaded repeatedly. AI-generated resources are clearly identified as AI-assisted rather than presented as original teacher material.

4.8 Workspace Navigation Map

- **Student path:** Student home (**VS-15**) → My Classes → Class 9-A workspace (**VS-09**) → Subject (Physics) → Chapter (Motion) → Concept list → Concept page: notes, session, recording, quiz.
- **Chapter resources:** notes (**VS-11**), recording segments (**VS-12**), curated video (**VS-13**), history (**VS-14**).
- **Also from home:** Class Space (**VS-10**) · Homework list (**VS-17**) → submission screen · Concept map view (**VS-21**) of the chapter.
- **Teacher path:** Teacher dashboard (**VS-60**) → Class → Chapter workspace → publish/manage resources (**VS-62** , **VS-63** , **VS-65**).

4.9 Workspace Checklist

- [ ] Every resource requires concept tags before publish (VS-11)
- [ ] Recording upload asks for segments; school permission honoured (VS-12)
- [ ] YouTube links store start/end timestamps (VS-13)
- [ ] Chapter content from day one is reachable all year (VS-14)
- [ ] Resource records carry source, status and permissions metadata
- [ ] Class Space has updates but NO free-form student chat

PART 5 · STUDENT

Student World: Dashboard and Learning

The student opens the app and sees one question answered above all others: what do I study now, and what do I do after?

VS-15 – VS-27

VS-15 Student Dashboard — "What Do I Study Today?"

CORE P1

Screen layout, top to bottom:

1. **NEXT ACTION CARD** (topmost, exactly one thing): *"Session 4 — Motion, 15 minutes. Start?"* The dashboard's job is to give DIRECTION, not guilt.
2. **TODAY'S WORK:** homework due, available sessions, tomorrow's quiz.
3. **CONTINUE CARD:** resume exactly where the student left off — one tap.
4. **MY CONCEPTS:** a small concept map (**VS-21**) — what is green, what is red.
5. **PRIORITY-AWARE REORDERING:** deadline today? That card rises to the top. Missed class? The recovery card (**VS-20**) takes the top slot. The dashboard is always dynamic, never crowded.

Blueprint dashboard sections (S5 §18) add: cloud companion personal welcome; upcoming assignments and tests; active guided sessions; continue-learning card; recently studied lessons; teacher announcements; learning progress by subject or concept; recommended practice; recent achievements or milestones; and a help/question-asking entry point. The dashboard prioritises actual learning tasks over decorative statistics.

OPEN: should the dashboard show all subjects at once, or only the subject with the nearest deadline? (Section 17.4)

5.2 Today's Learning Plan

A short, understandable plan built from school-assigned tasks, deadlines and available learning evidence. A plan may contain: (1) a teacher-assigned session (mandatory); (2) a short concept review; (3) a practice activity; (4) an unfinished assignment; (5) a suggested recovery activity where evidence indicates a gap (**VS-20**).

RULES

The app must visibly distinguish mandatory teacher-assigned tasks from optional AI recommendations. The AI must never silently change a teacher's deadline or replace an assigned lesson.

VS-16 Guided Self-Study Sessions — *the heart of the product*

CORE P1 ★★

A guided session is a **teacher-created, structured learning package** — not a PDF, not a video, something in between. Fixed session structure — 10 steps:

1 · Session intro
"What we will learn today".

2 · Learning objective
Stated plainly, before content.

3 · Explanation slides
ONE slide = ONE small idea.

4 · Diagram / visual
See the concept, not just read it.

5 · Solved example
Step by step, in the teacher's method.

6 · Media (optional)
Recording segment (**VS-12**) or trimmed YouTube (**VS-13**).

7 · Practice questions
3–4, inside the session.

8 · Mini quiz
5 MCQs — the completion gate (**VS-26**).

9 · Recap
What was covered, compressed.

10 · Next step
E.g. Speed & Velocity — where to go next.

Student flow: the session opens at the teacher-set time (e.g. 6 PM) → slides presented one at a time → questions INSIDE the session must be answered before moving on (no skipping ahead) → quiz → score recorded → next-step recommendation.

Teacher flow: Create Session → fill the template (or start from an AI-drafted skeleton, **VS-67**) → set "Available: today 6 PM" → publish → next year, press "Reuse" (**VS-68**): update dates, swap questions, done.

THREE NON-NEGOTIABLE DESIGN RULES

1 · Completion happens ONLY when the quiz is attempted — scrolling is not completion (**VS-26**).
2 · One idea per slide. 3 · Sessions run 15–20 minutes — evening time is limited.

The session player also records progress as the student works, allows clarification requests where enabled, and never inserts artificial timers just to inflate screen time.

5.4 Time-Controlled Lesson Release

- Teacher configures the release date and time; the system stores the correct timezone.
- Before release, the student sees a clear “upcoming” status.
- At release, the session becomes available to authorised students.
- Deadlines are configured separately from release times.
- Late access or extension rules follow school policy.

VS-17 In-App Homework & the Assignment Workflow

CORE P1

Teacher side: creates homework (MCQ / short answer / practice set) → sets deadline → assigns → students solve inside the app → MCQs auto-grade → teacher gets the list: who did it, who did not, who was late.

Student workflow (S5 §24): open the assignment → read instructions → review the relevant material → submit answers/work → receive submission confirmation → review feedback once released → complete corrections or follow-up if assigned.

Assignments can include: instructions; questions; supporting resources; due date; submission type; marks or rubric; allowed attempts; feedback and review status.

IMPORTANT BOUNDARY

Writing does not end. Derivations, diagrams and handwriting remain pedagogically necessary. Only repetitive, checkable work goes digital — **the TEACHER decides** what is online vs in the notebook. Assignment completion and concept mastery are recorded as SEPARATE things.

VS-18 Post-Session Quiz + “Why Was It Wrong?” Feedback

CORE P1

1. Student submits the quiz — e.g. 4/5.
2. Question 3 was wrong → feedback: *“Question 3 was about the concept of Acceleration.”*
3. A button appears: **“Reopen the explanation.”**
4. The student re-reads that exact slide.
5. **“Try again”** presents a SIMILAR NEW question (not the same one).

Why it is a real feature: normally a quiz result goes nowhere — marks are received and forgotten. This closes the loop: mistake → reason → fix → retry. Assessment exists to decide what happens next, not only to assign marks (S5 §4).

VS-19 Two-Line Summarizer

PROPOSED

A button on any teacher material: the AI summarises THAT material into 2–3 simple lines, with the option to go deeper. The source is the teacher's material, not the internet — this is what makes it different from ChatGPT. Best for orientation: "What is this topic about?"

VS-20 Missed-Class Recovery Path

CORE P2 ★

Triggers: the student was absent (school record), or missed a deadline, or failed a prerequisite check. The recovery path is assembled from EXISTING teacher material — no new content is created:

1. Dashboard card: *"You were not in class yesterday. Catch up in 20 minutes?"*
2. The 7-minute recording segment of the missed concept (**VS-12**).
3. Teacher's notes (**VS-11**).
4. A mini session (**VS-16** structure).
5. A 5-question check.
6. **PASS** → back at class pace. **FAIL** → the Gap Detector (**VS-30**) searches for the root cause, one prerequisite level deeper.

Why it is emotional: a missed class becomes a small, fixable event instead of a silently accumulating gap. The founder's own story broke on exactly this cascade — this feature stops it. The Blueprint calls this the *Learning Recovery Engine* (S5 §52) and specifies it must help students catch up WITHOUT repeating an entire chapter unnecessarily; the teacher can see recovery progress where appropriate.

VS-21 Visual Concept Map

PROPOSED

A tree view of the chapter. Nodes are concepts; **colour = the student's status**: *Distance OK — Velocity OK — Acceleration ATTENTION — Graphs LOCKED.* The student sees for themselves: "Acceleration is red — and everything after it depends on this. Fix this first." This is metacognition: seeing your own map and planning from it. Locked nodes unlock as prerequisites are demonstrated (linked to **VS-100** Boss Level).

VS-22 Four Performance Bands — internal only

PROPOSED

Band	Internal meaning (never shown as a label)	Support offered
0–25%	Large gaps	Simplest explanation, prerequisite revision
25–50%	Basics partly clear	Worked examples, guided practice
50–75%	Basics clear	Application questions, more practice
75–100%	Consistent understanding	Next concept, harder challenges

Two non-negotiable rules: (1) bands are NEVER shown to students as labels — students see friendly language: "Mastered", "Almost there", "Let's fix this". (2) Bands are a ROUTING MECHANISM, not a

scientifically validated measure. People are not labelled — CONCEPTS are: “Rahul — Distance: 82%, Acceleration: 30%.” A single overall percentage hides WHICH concept is weak; per-concept tracking is probably better, and thresholds are not validated (OPEN — Section 17.4).

VS-23 Adaptive Difficulty Engine

PROPOSED

Every question carries a difficulty tag: **L1** basic, **L2** medium, **L3** application. Rules: student starts at L2 → 3 consecutive correct → move up to L3 → 2 wrong → drop back to L1 with an explanation. Simple rules, no magic. **Payoff:** the student is neither bored nor crushed — they stay in the zone where effort is visible. This engine underlies the practice engine (**VS-46**) and both games and quizzes.

VS-24 Exam-Format Practice

PROPOSED

Every question in the bank carries a FORMAT tag: MCQ, Assertion-Reason, Case-Based, Numerical, Short Answer — the real board exam patterns. Before exams, a “Board-format practice” mode shows exactly what will appear in the exam. Knowing a concept and practising it in board format are two different skills.

VS-25 Test Readiness Signal

PROPOSED

Before a chapter test: *“Strong: 4/6 concepts. Attention: Acceleration. Suggested: 30 minutes of focused revision tonight.”* Computed as a weighted average of per-concept scores and recent history — simple math. **What it is NOT:** it never predicts marks (“You will score 62%”) — predicting marks would be a claim of determinism. Only “these areas are weak.”

VS-26 Session Completion Rules

CORE P1

- Slides unlock sequentially, one at a time.
- A session is complete ONLY when its quiz/check is attempted (the teacher sets this rule).
- Opening a slide is NOT learning — a check is mandatory.
- Addresses the known failure modes: opening all slides at once (looks like progress, learns nothing), skipping the test (no evidence of understanding), closing the app mid-work (short sessions, saved progress, gentle reminders).

VS-27 Student Progress View & Learning History

CORE P1

Progress view: what is complete, what is pending, mastered concepts, this week’s improvement (“+15% on Acceleration!”), streak (**VS-93**) — and above all ONE clear next action. The dashboard gives direction, not guilt.

Progress dashboard sections (S5 §26): progress by subject; chapters studied; concepts demonstrated; concepts needing more practice; assignment status; assessment history; recent improvements; recommended next steps; teacher feedback; learning passport or skill evidence where enabled. The student is never reduced to a permanent label.

Learning history (S5 §27): lessons opened and completed; questions attempted; assessment results; repeated error patterns; teacher feedback; learning plans and interventions; evidence of improvement

over time. Only information needed for educational purposes is retained, with access and retention controls (Part 13).

5.16 Notes, Videos, Assessments and Language Preference

- **Notes & resources (S5 §22):** students access teacher notes, school PDFs, worksheets, presentations, diagrams, recorded lessons where available, approved external resources, and AI-generated study material when enabled and reviewed per school policy. Every resource shows title, subject, chapter and source. AI-generated resources are clearly identified.
- **Video learning (S5 §23):** teacher attaches a video with a relevance explanation → student watches → completes a short check tied to the learning objective → the app records the activity separately from verified mastery. *Opening a video is never counted as understanding.*
- **Assessments (S5 §25):** multiple-choice; multiple-select; short-answer; numerical; matching; teacher-created question sets; timed tests where educationally justified; answer explanations; rubric-based evaluation. Teachers control when correct answers/feedback become visible. For high-stakes assessments, AI output never independently determines final grades without human review.
- **Language preference (S5 §28):** English, Hindi, or Hinglish-style explanations (**VS-90**). Curriculum and examination terminology must remain accurate: an informal explanation can be followed by the standard scientific definition expected in the exam.

5.17 Student Screen Connection Map

Blueprint student screens (S5 §87), in workflow order: Home dashboard → Today's plan → Class & subject view → Chapter & concept view → Guided session player → Resource viewer / Video page → Assignment details → Assignment submission → Quiz & assessment player → AI question interface → Personalised learning plan → Learning Remix → What-If simulation → Mistake analysis → Progress dashboard → Learning history → Learning Passport → Notification centre → Profile & preferences. (The last screens belong to the advanced roadmap rather than the initial prototype.)

DEEP-LINK RULE

EVERY red/attention state anywhere in the app leads to the screen that fixes it — the concept map node taps into the session; the quiz feedback button reopens the explanation slide (**VS-18**); the recovery card starts the recovery path (**VS-20**).

5.18 Student-World Checklist

- [ ] Dashboard always shows exactly ONE next action, dynamically reordered
- [ ] Sessions follow the 10-step structure; one idea per slide
- [ ] In-session questions block progression until answered
- [ ] Quiz attempted = completion; scroll-only never completes (VS-26)
- [ ] Wrong answers route: reason → reopen explanation → new similar question (VS-18)
- [ ] Recovery card appears automatically on absence/deadline-miss triggers
- [ ] Bands never visible as student-facing labels
- [ ] Difficulty engine routes L1/L2/L3 per 3-up/2-down rules
- [ ] Mandatory teacher work visually distinguished from AI suggestions

- [ ] Video watching and activity completion recorded separately from demonstrated mastery

PART 6 · AI ENGINE

The AI Engine

Every AI feature here is checked against one question: **"Can ChatGPT already do this?"** If yes — it is not a moat. The features below pass the test because their fuel is data only a school-connected platform has: the teacher's notes + the class's quiz results + the curriculum position + the student's behaviour log.

VS-28 – VS-59 + BLUEPRINT AI MODULES

6.1 AI Foundations: Grounding and the Honesty Layer

VS-32 AI Honesty Layer — *rules above every AI feature*

CORE

1. **CITE THE SOURCE:** answers distinguish "from your teacher's notes" vs "general explanation."
2. **ADMIT THE UNKNOWN:** when the AI does not know, it says so — no confidence theatre.
3. **CAUTION OUTSIDE MATERIAL:** factual claims beyond the teacher's material are flagged as general knowledge, not school answers.
4. **SENSITIVE TOPICS:** the AI does not reply; it routes the student to the teacher.
5. **NO INVENTED AUTHORITY:** the AI never says "your teacher said" when it does not know.

Trust is the AI's foundation — for the child AND the teacher. The Blueprint (S5 §78) formalises this as the *Teacher-Approved Knowledge Layer*: prefer relevant school-approved sources; identify the source used; distinguish source-supported facts from generated explanation; say when an answer is not supported by available material; allow teacher corrections to flow back into approved content.

6.2 Core Intelligence (VS-28 to VS-34)

VS-28 Context-Grounded Ask-Me-Anything

CORE P2 ★

The student asks *"explain acceleration"* and the AI already knows the context: Class 9-A, CBSE, Motion chapter in progress, THESE teacher notes exist, this student got Acceleration wrong on yesterday's quiz. The answer is delivered in the **TEACHER'S OWN NOTATION**, ending with the source: *"This is from your teacher's notes — Motion, slide 4."* **Guardrail:** if the material does not contain the answer, the AI says "I'll give you a general explanation" — so the student always knows what is school-backed and what is not. The AI never claims to have read something that was never uploaded (S5 §46). **Why it passes the moat test:** ChatGPT claims to know everything. This AI lives inside the school's reality.

VS-29 Socratic Mode — *the homework-copy killer*

CORE P2 ★

The problem: "The child will just make ChatGPT do the homework" — a legitimate fear. The design-level answer:

STUDENT
"Solve this numerical for me."

AI
"First tell me — what is given, and what must be found?"

STUDENT
"Distance and time are given; velocity must be found."

AI
"What is the formula for velocity?"

STUDENT
"...displacement by time?"

AI
"So what do you need for displacement?"

The AI asks only path-guiding questions; the STUDENT arrives at the answer alone. A "just explain it directly" toggle also exists; homework-type questions can default to Socratic mode (teacher setting). One feature answers the entire "AI makes students dumb" criticism.

VS-30 Prerequisite Gap Detector

CORE · P2 · ★★

What a good teacher does: when a child gets acceleration wrong, first checks "is speed-velocity okay though?" This feature puts that teacher-logic into code. **Mechanics:** the syllabus carries a concept map: Distance → Speed/Velocity → Acceleration. Student B gets acceleration wrong → system checks: were the velocity questions answered correctly? YES → acceleration revision. NO → step one level back → velocity also wrong → step back again → all the way to Distance if needed. Wherever the chain broke, it stops there: *"Velocity is a little weak before acceleration. Fix it in 10 minutes? Acceleration will feel easier afterwards."*

Guardrail: if evidence is missing (the student never attempted velocity questions), the system does NOT guess: "Practise, or ask the teacher." An honest "I don't know" is also a feature. This is the real work of coaching/tuition — going backwards to find the root problem; the feature scales the teacher's clinical thinking. (How many questions of which types are enough evidence before suggesting a prerequisite gap is OPEN — Section 17.4.)

VS-31 Study Plan Generator

CORE · P2

Input: the student says "I can't understand Coordinate Geometry." The system checks the student's results, then builds: (1) prerequisite check → (2) the teacher's notes — these specifically → (3) the 6-minute recording segment on plotting points → (4) one solved example → (5) 3 easy questions → (6) a 5-question check → (7) pass = application questions; fail = return to step 3. **What is special:** every step links to the school's ACTUAL resources — not a generic timetable but a path that runs through school material and re-routes on results. Teacher-assigned work stays visible and takes precedence (S5 §51).

VS-33 Frustration-Aware AI

PROPOSED · P2

What it is NOT: no camera/microphone emotion detection — creepy with children, a privacy nightmare, deliberately cut from the design. **What it IS:** behaviour signals — the same question asked 3 times, 5

wrong answers, then the same question again → the pattern shows the student is stuck → the response changes: *"It's okay — this one is a bit tricky. Let's start from absolute zero, the very basics. Tell me, what have you understood so far?"* Tone changes, pace changes. No biometrics — only interaction patterns.

VS-34 Warmup Radar — the Forgetting Detector

PROPOSED

Before every session, a 3-minute warmup — not random questions: the AI picks exactly the concepts THIS student is about to forget. Every concept has a record: last attempted when, score how much → simple forgetting-curve math → score + time-since-seen → "this is due for recall now" → warmup queue. **Payoff:** spaced repetition for free from the school's own quiz data — the student never has to schedule revision separately; it is baked into warmups.

6.3 Doubt Engine (VS-35 to VS-38)

VS-35 Voice Doubt Asking

PROPOSED P2

Press the microphone, speak in Hindi or Hinglish, hear the answer. Speech-to-text → the normal AI flow (VS-28) → answer as text + optional audio. **Why India-critical:** many students — Tier-2/3, younger ones — cannot type fluently. Typing Roman Hindi feels awkward; English typing is hard. Speaking takes 3 seconds; typing takes 30 seconds plus embarrassment.

VS-36 SnapDoubt — Ask by Photo

PROPOSED P2

The student photographs a notebook page → the Vision API recognises *"this is an Acceleration numerical (Motion, Ch.2)"* → teacher's notes enter the context → Socratic questioning begins (VS-29). Three API calls, that's all. **Different from ChatGPT:** (a) it uses the TEACHER'S method — the one that earns step marks in the exam; (b) it knows the student's history; (c) Socratic mode is the default. Exam-scoring answers, not just correct answers. **Fallback:** a "type it instead" button — younger children's handwriting can be rough.

VS-37 Doubt Clustering — *the teacher's secret weapon*

PROPOSED P2 ★

Scene: at night, 17 children asked about acceleration in 17 different wordings. Every AI question is auto-tagged with its concept → the teacher's dashboard next morning reads: *"This week: 17 students asked doubts on Acceleration. 3 on Velocity. Zero on Distance."* **Guardrail: NAMES ARE NOT SHOWN BY DEFAULT — counts only.** If children learn the teacher can see WHO asked, they stop asking doubts, and the entire AI layer becomes useless. Doubt privacy = doubt volume. Zero extra teacher effort, pure insight — from the doubts that were flowing anyway. (Names vs counts-only is OPEN — Section 17.4; the default is anonymous.)

VS-38 Doubt Inbox — opt-in, named

PROPOSED

Student not satisfied with the AI's answer → "Ask the teacher" button → the doubt lands in the teacher's inbox WITH the student's name → the teacher prioritises and answers. **Principle:** the AI does not replace the teacher — it FILTERS for the teacher. Basic doubts are absorbed by the AI; real problems reach the teacher.

6.4 Psychology Engine (VS-39 to VS-42)

VS-39 Confidence Meter *("Pakka Meter")*

PROPOSED | P2 | ★★

Before every question, three buttons: **"Sure" / "I think" / "Guess"**. The system remembers the confidence AND the outcome. Two patterns nobody tracks:

SURE + WRONG = OVERCONFIDENT

The most dangerous silent gap. *"You were sure — and it was wrong. Revise Acceleration seriously."*

GUESS + RIGHT = UNDERCONFIDENT

The child knows, but is afraid. *"You actually know this! Relax, don't panic."*

Weekly: "You are overconfident in 6 concepts, underconfident in 2." The teacher also sees who is quietly getting things wrong. **Why:** metacognition — knowing your own mind — is one of the most validated skills in learning, and this market space is literally empty. Implementation: one extra field + one query. (Guardrail: self-reported confidence is never treated as proof of mastery — S5 §69.)

VS-40 Mistake DNA *("Galti DNA" — error fingerprinting)*

PROPOSED | P2 | ★

Principle: "It was wrong" is information. "WHY it was wrong" is gold. The AI classifies every wrong answer:

Type	Meaning	Fix
Careless	Concept known; sign/calculation slip	Slow-down checklist
Concept	Understanding missing	Reopen notes; prerequisite check (VS-30)
Speed	Knew it; rushed	Timed practice
Reading	Misread the question	Underline habit

Monthly DNA report: *"60% of your mistakes are CARELESS — you don't need more study, you need a HABIT fix. Here is a 30-second checklist."* This answers the parents' #1 confusion ("why are the marks low?"): "It's not a concept problem, it's a focus problem." Two different problems, two different solutions — which until now look identical to every parent. The Blueprint's Homework Autopsy (S5 §62) lists the same classification set — each treated as a possible explanation, not a diagnosis.

VS-41 Rote-Learning Detector *("Ratta Detector")*

PROPOSED | ★

India's biggest silent learning disease: definition perfect, marks arrive, application = zero. Detection uses CROSS-FORM questioning: (1) "Define acceleration" — rote can pass this; (2) "A cycle is braking — what is the sign of acceleration?" — rote cannot pass this; (3) "Compare the acceleration of a fast bowler's ball vs a school bus" — reverse direction. **Score mismatch = rote flag:** *"Definition 5/5, application 1/5 — this looks like rote learning. Let's reopen the concept, 7 minutes."* **Teacher view:** "14 students have perfect definitions; 9 have weak application — this is the 'everyone understood' illusion that a normal test report never catches."

VS-42 Avoidance Radar *("Darr Ka Map" — the fear map)*

PROPOSED ★★

The deepest insight in the spec: a child who cannot understand a concept does not answer it wrong — they AVOID it. No app tracks this, because no app keeps this data. VidhyaSaathy keeps it: least time spent on Acceleration questions (40% less than others); 6 skips on that concept in 4 weeks; instantly requests hints on it, but not on other concepts; attempts the quiz but never voluntarily reopens that chapter.

AI INTERPRETATION

"The data suggests you've been avoiding Acceleration — for 3 weeks. Completely normal: everyone has 1–2 feared concepts. Today's deal: just 3 questions, no hints. I'm with you." — normalise, never shame. Everyone tracks "wrong answers" — nobody tracks "avoidance patterns across weeks", because it needs both a behaviour log AND school context. This is the silent killer that explodes two weeks before the exam; the system catches it months earlier.

6.5 Practice Engine (VS-43 to VS-46)

VS-43 AI Question Generation — teacher-approved drafts

PROPOSED P2

Teacher selects their own Motion notes → "Generate 10 MCQs" → AI drafts → teacher approves 7, edits 2, rejects 1 → approved items enter the school's question bank (VS-87). **Non-negotiable:** AI-generated content NEVER reaches a student directly. AI can produce a wrong question or answer key, and wrong facts are expensive for children. AI = assistant; teacher = authority.

VS-44 Interest Remix

PROPOSED P2 ★

During onboarding the student picks interests — cricket, bikes, cooking, K-pop. The AI remixes questions: SAME concept, SAME difficulty, SAME marking scheme, only the wrapper changes:

NORMAL

"A car accelerates from 0 to 20 m/s in 5 s. Find the acceleration."

REMIX — CRICKET FAN

"A fast bowler's ball goes from 0 to 140 km/h in 0.4 s. Find the acceleration. (Hint: convert to m/s first.)"

The physics is the same, the formula is the same, the exam will ask the same thing — but the child practises inside their own world, and shares screenshots (free marketing by the students themselves). **Safety:** teacher sample-check flow (spot-check 1 in 10) — full per-question approval would exhaust the teacher. A report button exists for weird or embarrassing remixes.

VS-45 Personal Enemy Questions *("Dushman Sawaal")*

PROPOSED ★

A new question arrives → embedding similarity check → **"STOP. This is your enemy. You got this exact type wrong on 14 October."** → battle mode: enemy list ("You have 5 enemy questions — defeat 2 today"); defeat an enemy by answering it correctly **3 TIMES IN DIFFERENT FORMS** → confetti, badge, strike-through. The list updates after every quiz — new enemies appear, defeated ones are crossed off. **Why it works:** no child keeps a record of their own repeated mistakes — but everyone remembers their ENEMY LIST. Spaced repetition turned into a personal war.

VS-46 **Adaptive Difficulty — practice link** PROPOSED

The routing core from **VS-23** (L1 → L2 → L3, 3-correct-up / 2-wrong-down). The Interest Remix and Personal Enemy systems both run on top of it. The Blueprint's adaptive practice (S5 §66) adds the action menu: another example of the same skill; a changed question representation; a return to a prerequisite; a harder problem; a hint; or teacher support — with recommendations explained when useful.

6.6 Explanation & Companionship (VS-47 to VS-50)

VS-47 **Hinglish Bridge** — *English notes → Hinglish understanding* PROPOSED P3 ★

India's age-old tension: notes are in English (textbook/board English), but the child THINKS in Hindi/Hinglish. That gap is the real wall to understanding. A toggle on any English material — "Explain in Hinglish" — the AI re-explains the SAME content with technical terms kept in English:

HINGLISH BRIDGE SAMPLE

"Here's the scene — Displacement means your STRAIGHT-LINE distance from start to end, including direction. Distance means everything you travelled, the total path. Simply: if you moved around, distance grew, displacement stayed the same."

What coaching really sells is not content — it is TRANSLATION. "The same textbook thing, in your language." This feature delivers that bridge without coaching, from the teacher's own material — same notation, same method that earns exam marks.

VS-48 **"Show You Understand" Mode** *(Feynman, "Samjha Ke Dikhao")* PROPOSED ★

"Explain acceleration as if to a friend." The student speaks (voice) or writes → the AI plays teacher and listens: *"Three things right. One missing: the role of direction. You said negative acceleration means the opposite direction — not quite. Look at this: the teacher's recording, 2:10–2:50, just 40 seconds — then try again."* Inside: speech-to-text → LLM rubric grading of the explanation → the exact recording segment + timestamp from the concept map. A generic AI can do Feynman practice; only VidhyaSaathy can point to WHERE in the teacher's recording THAT point lives. The protege effect: what you teach someone else is your most durable learning.

VS-49 **Night Recap** *("Raat Ka Recap" — personal 5-minute podcast)* PROPOSED P4

The day's data becomes a 9 PM notification: *"Today's recap is ready — 5 minutes. Listen, then sleep."* Conversational Hinglish audio: *"So today you learned distance vs displacement. 4/5 correct? The one wrong was acceleration — once more: acceleration means the change in velocity per second..."* Content is exactly three things — what was learned today + the fix for today's mistake + a 30-second preview of tomorrow. Personalised: the student's own quiz numbers, own mistakes, own language. "The school's own radio channel, just for you." On a shared family device, the parent overhears it too — indirect parent awareness.

VS-50 **AI Big Brother / Big Sister** *("AI Bhaiya-Didi")* PROPOSED ★★

Insight: India's most common tutor is not a teacher — it is the elder sibling who passed through the same board, same school, same stress two years ago. The AI sibling is exactly that: voice-based,

casual, a "been-there" tone:

STUDENT
"Chemistry just isn't going into my head at all..."

AI DIDI
"Yeah, Class 9 chemistry feels the most boring at first. I almost cried over it too. Know what I did? Come on, today just 20 minutes — only periodic trends. I'm with you. Tomorrow I'll tell your mom you started."

It knows the Confidence Meter results and the Enemy List — but speaks like an elder sibling, never like a teacher or an AI. **The TONE is the trust.** Weekly parent nudge: *"Aarav completed 4 sessions this week. Wonderful. Give him a small treat on Sunday — tell him I sent this message."* A family-communication layer no other product has.

6.7 Simulator Lab (VS-51 to VS-54)

VS-51 At-Home Viva *(AI oral exam, "Ghar Baithe Viva")*

PROPOSED · P4 · ★

India-specific: viva examinations are real — practicals, boards, internals. Viva fear is its own category: students can write but cannot SPEAK, and there is nowhere to practise. No teacher can conduct 60 individual vivas. *"Viva Mode"* → the AI becomes a teacher (voice):

AI
"Define acceleration."

STUDENT
"Change in velocity... per unit time."

AI
"Correct. And what does negative acceleration mean?" ← *the follow-up: THIS is the real viva*

End: *"Viva score 6/8. Theory is solid; definitions stumble — check the flashcards, 2 minutes."* Questions come from teacher material + board viva patterns. A text fallback must exist so a live demo never fails on stage.

VS-52 Exam Hall Simulator — *the pressure vaccine*

PROPOSED · P4 · ★

What no edtech addresses: the concept is known — but in the hall it vanishes. Blank mind, trembling hands, clock panic. That is not a knowledge problem, it is a PSYCHOLOGY problem. Athletes get simulation training; why not students? Full sensory exam experience: real board-paper format + a ticking 3-hour clock + background audio (paper rustling, bells, the invigilator: *"Fifteen minutes remaining"*) + RANDOM BLANK-MOMENT EVENTS → *"That feeling is coming? Breathe 4-7-8. 30 seconds. Come back."* — practising the recovery protocol itself. **Report:** time-block analysis — "Only 2 questions in the first 30 minutes: panic pattern. Strategy: pick easy targets first." **The parent line:** "74% in the simulated hall, 85% at a calm table — 11 marks lost purely to pressure. THIS is what we train."

VS-53 **Reverse Class** — *the AI plays the weak student* PROPOSED ★★

The world makes AI the student's tutor. HERE THE AI PLAYS THE STUDENT — and deliberately makes REAL mistakes:

> AI
> *"Sir, acceleration means speed increases, right? So if the car is slowing down, acceleration must be zero?"* ← *wrong — the child must catch it*
>
> STUDENT
> *"No! Slowing down means acceleration is NEGATIVE—"*
>
> AI
> *"Oh?! So the car stops the instant brakes are applied? How long does negative acceleration last?"*

The AI's errors are real: they are the same misconceptions found by the class-level autopsy (VS-70). Whatever misconception is spreading in the class gets planted inside the AI student for the child to catch — catching the AI's error removes the misconception from the child's own head. Children ENJOY this — the pleasure of correcting someone. (Blueprint territory: "AI Debate Between Past and Future Thinking", S5 §61 — using the student's earlier wrong reasoning as the thing to argue against.)

VS-54 **Digital Twin** — *tomorrow's result, today* PROPOSED P4 ★★

From the student's current mastery, the AI composes a SIMULATED next test paper — the "Twin" — and lets the student attempt it today: *"Twin Result: 58/100 — at today's level. Acceleration will betray you (62% chance of wrong). Now the choice is yours: 2 sessions tonight → 71. Skip them → 58."* **The motivation cheat code:** "Study or else..." never works — it is abstract. "58 vs 71, in your hands" is concrete, personal, immediate — consequences time-travelled into the present. Parent version: "Simulated 58. Potential 71. The gap comes down to 1 concept — being fixed tonight."

6.8 Time-Machine Layer (VS-55 to VS-59)

VS-55 **"Teacher Said So" Radar** *("Sir Ne Bola Tha")* PROPOSED ★★

India's most powerful unspoken phrase: every topper's secret is knowing WHAT THE TEACHER STRESSED. "This WILL be asked" — that sentence wins classrooms. Until now, only the students who sat attentively had it. Recording uploaded → the AI hunts EMPHASIS MOMENTS: where the teacher said "this is important, remember, I will ask from this"; where ONE thing was repeated 3 times; where the most doubts came; even where a joke landed (humorous things are remembered). **Student feed:** *"6 gem moments from yesterday's class: (1) 'uniform motion' stressed 4 times; (2) 'careful with sign convention' — a direct warning; (3) 3 children had the same doubt about negative acceleration..."* **Teacher Mirror** (VS-71): "You spent 12 minutes on acceleration, interaction only 2 minutes. Yesterday's quiz missed that concept the most. Connection?" **Uncopyable:** the recording the SCHOOL uploaded + that teacher's voice.

VS-56 **Mistake Autopsy** — *thinking-path reconstruction* PROPOSED ★

Every app says "it was wrong." This feature finds the exact moment the thinking broke — detective, crime-scene style: *"Let's look: Step 1: correct formula chosen — OK (concept known). Step 2: '12'*

side of the Blueprint's Mistake Time Machine, S5 §54: parse the steps the system can confidently interpret → compare each step with the relevant rules → highlight the EARLIEST likely incorrect step → explain why → give a similar question → check the fix. Ambiguous handwriting triggers a confirmation question, never a guess.)

VS-57 **Teacher DNA** — *the data science of "what will be asked"* PROPOSED ★★

Every school's ritual: ask a senior — "what does this teacher ask?" The seniors hold a 3-year pattern in pure gut feel. India's oldest prediction system becomes data science here. The teacher's 3–5 years of papers + this year's quizzes + emphasis signals from recordings → a Teacher DNA profile: *"Mrs. Sharma's 5-year pattern: 68% application questions; definitions never above 1 mark. Graph-based questions every year (5/5). 'Uniform motion' stressed 4x in recordings. This time HIGH priority: graphs, uniform/non-uniform motion, numericals."*

> **ETHICS LINE (DELIBERATE)**
> This is NOT a guess-paper. It organises the teacher's OWN emphasis signals. THE TEACHER SEES AND APPROVES THE PROFILE FIRST: "This is your pattern — show students this priority?" Only when the teacher says yes. The teacher's wisdom, structured — nobody else has a teacher's 5 years of papers.

VS-58 **Syllabus GPS** PROPOSED ★

The Google-Maps feeling: **LOCATION** — Class 9-A is here: Speed & Velocity. **YOU** — you are here: Distance & Displacement (2 back). **DESTINATION** — Board exam: 14 chapters | 6 months. **AI route:** *"You are 2 concepts behind. The route: Acceleration recording (7 min) + 10 questions + 1 revision — 20 minutes/day, level in 4 days. The traffic jam is at Acceleration — that's exactly where you kept skipping."* EVERY QUIZ RECALCULATES THE ETA — like Maps rerouting after a wrong turn. Class position + student position + exam date: this triangulation needs a school-connected platform. The pitch's hero visual: *"Google Maps, but for the syllabus."*

VS-59 **PTM Time Machine** PROPOSED ★★★

The PTM — India's most awkward 6-minute ritual. The parent hears "needs improvement," goes home, scolds the child. Nobody gains. THIS FEATURE REHEARSES THE MEETING BEFORE IT HAPPENS. **Parent mode:** "PTM is tomorrow. Practise?" → a simulated teacher built from the CHILD'S REAL DATA:

> **AI (AS TEACHER)**
> *"Aarav is a little behind in Physics..."*
>
> **PARENT**
> *"What should I do?"*
>
> **AI**
> *"That was instinct. Try instead: 'Where exactly is he getting stuck in acceleration?' — ask specifically, and the teacher will answer specifically."*

The parent is trained to ask EXACTLY 3 data-backed questions. **Student mode:** the child watches the simulation — "Tomorrow Ma'am will say: quiz 2/5. Your response?" — and prepares: *"Ma'am, I did the*

recovery session; now it's 4/5 — shall I show you?" The student walks into the PTM with their own growth story, defended with evidence. Not defensive — evidence-based. Nobody in the world has made the PTM a product feature.

6.9 Blueprint AI Modules (beyond the VS feature list)

The Blueprint (S5 Part 8) specifies additional AI modules, preserved with their citations:

Module	What it specifies
BP-§47	**Personalised Explanation Engine** — the AI switches representation: simple language; step-by-step; everyday analogy; visual walkthrough; worked example; formal textbook explanation; exam-answer format; bilingual. After explaining, the system CHECKS whether understanding improved; if not, it tries another approach or recommends the teacher — never endlessly rephrases.
BP-§48	**Learning Remix** — the same concept as mystery, counterexample, everyday analogy, visual, challenge, guided example, or short story — always linked to the learning objective and school-approved knowledge.
BP-§49	**Student Learning-State Model** — per concept: introduced or not; evidence from attempts; accuracy across question types; repeated error patterns; recency of practice; confidence in the system's own estimate; teacher-confirmed status; recommended next action. A working educational estimate — never a measure of intelligence.
BP-§50	**Concept & Prerequisite Graph** — concept chains (arithmetic → rearranging equations → substitution → speed → acceleration), prerequisite checks when a later concept struggles; teachers can review or correct curriculum relationships.
BP-§53	**What-If Machine** — interactive scenarios ("What happens to speed if distance increases but time stays constant?") with DETERMINISTIC calculations for numbers; stated assumptions and limits; AI only EXPLAINS results, never invents the simulation; students PREDICT before the reveal; ends with a check question.
BP-§55	**Chapter-to-Comic Studio** — an approved concept becomes a 4–6 panel illustrated explanation: curriculum-grounded story, dialogue + concept notes, reusable character poses, science verified, concept check at the end.
BP-§56	**Infinite Textbook Adventure** — a branching learning story on a curriculum concept; correctness decided by predefined educational rules; scoring never decided by unconstrained AI text; ends with summary + practice. Initial scope: ONE short, carefully tested adventure.
BP-§57	**Concept Court** — two fictional characters argue competing claims about a school concept; the student decides which claim is better supported, which reasoning step is wrong, what evidence settles it, and how to state the conclusion.
BP-§58	**Reverse Exam Mode** — the student writes an answer, receives rubric-based feedback, reviews a suggested improvement, REWRITES, and compares original vs revised. AI feedback is formative; teacher grading stays authoritative.
BP-§60	**Reality Scanner** — connects curriculum concepts to everyday objects: a bicycle → motion; a kitchen measurement → ratios; a plant → biological structures. Camera recognition is a LATER possibility, not assumed.
BP-§63	**Parent Translator** — converts learning evidence into parent-facing language (see Section 8.2).

BP-§64 **Learning Passport** — a record of demonstrated skills with evidence, teacher verification, status + next step; could later support school-to-school moves with permission. Never an unchangeable profile of a child's ability.

6.10 AI Request Workflow, Governance and Safety Rules

A typical school-related AI request follows this server-side sequence (S5 §93):

1. Authenticate the user.
2. Verify the user's permissions.
3. Identify the relevant school, class, subject and concept.
4. Retrieve relevant approved resources.
5. Include only the necessary learning context.
6. Send the request to the configured AI service.
7. Validate the returned structure and content where possible.
8. Attach source references when supported.
9. Save only the information required for the learning workflow.
10. Return the response to the user.
11. Record the educational outcome if a follow-up activity is completed.

AI responses are never trusted merely because they are well written.

> **AI SAFETY RULES (S5 §83)**
> Ground school answers in approved sources where possible · show uncertainty when evidence is insufficient · never fabricate citations · keep AI suggestions visually separate from teacher-approved content · no medical or psychological diagnoses, no permanent student labels · safe, age-appropriate explanations · protect student data from unnecessary disclosure · treat prompt injection inside uploaded documents as untrusted input · provide a fallback whenever the AI service is unavailable.

6.11 AI Engine Checklist

- [ ] Every AI answer carries a source tag: teacher material vs general
- [ ] "I don't know" paths implemented (VS-30 evidence-missing case)
- [ ] Socratic mode default on homework-type questions (teacher setting)
- [ ] Confidence buttons logged with every answer (VS-39)
- [ ] Mistake classification taxonomy (4-type DNA) wired into feedback
- [ ] Doubt clustering anonymous by default; names only via VS-38 opt-in
- [ ] AI question drafts ALWAYS pass through teacher review (VS-43)
- [ ] Interest remix respects: same concept, same difficulty, same marks
- [ ] Simulations use deterministic math, AI only explains (What-If rule)
- [ ] Fallback responses exist for every AI-dependent screen

PART 7 · TEACHER

Teacher World

One teaching effort, many uses — notes + recording + session + quiz from the same lesson. Teachers stay in control of content and assessment; the interface fits existing teaching habits and reduces work, never adds an administrative burden.

VS-60 – VS-72

VS-60 Teacher Dashboard — the morning class-plan decider

CORE P1

On screen: last night's session completion %, quiz score distribution, the Concept Heat-Map (VS-61), and the list of students who have not started. **The real magic — the insight-to-action loop:** *"Acceleration is red, 60% of the class got it wrong"* → teach a 10-minute reteach segment today → retest → heat-map turns green. Reactive teaching becomes proactive.

Blueprint dashboard sections (S5 §29) add: assigned classes and subjects; today's teaching and self-study sessions; draft and published lessons; assignments awaiting review; upcoming tests; class-level concept understanding; student questions requiring attention; common misconceptions; recent student progress; and quick actions (upload material, create a session, assign work).

VS-61 Concept Heat-Map & the Classroom Concept Radar

CORE P1

Heat-map: every concept of the class in one grid; colour = the % of the class that answered correctly — *green* (secure) / *peach* (mixed) / *red* (stuck). Acceleration red → the teacher understands instantly. **The second loop — catching it BEFORE class:** doubt clustering (VS-37) + quiz data show the gap before the next lesson. This is the data version of coaching's "personal attention" selling point.

Classroom Concept Radar (S5 §36): (1) students complete a diagnostic activity; (2) the system maps responses to known answer patterns / misconceptions (many confuse distance and displacement; some pick the wrong formula; others calculate correctly but cannot explain the result); (3) the dashboard shows the FREQUENCY of each pattern — concept-level view, not a class average; (4) the teacher reviews the evidence; (5) the teacher chooses: reteach, assign practice, or move on; (6) a follow-up check tests whether understanding improved. The radar must show uncertainty where a wrong answer could have several explanations.

VS-62 Publishing Notes & Resources

CORE P1

Concept-tagged upload (VS-11): a few taps — class → subject → chapter → concept → publish → notification to the class. **Rule:** notes published from DAY ONE of the chapter — students get the app aligned to the syllabus from the start, not months later. A teacher can also attach curated external videos (VS-13) and recordings (VS-12) where permitted.

7.4 Lesson Creation and the Lesson Editor

Lesson creation from existing material (S5 §30): supported inputs include PDF notes, text documents, presentation files, worksheets, typed lesson text, approved external links, and teacher-created questions. Workflow: (1) teacher uploads or selects material; (2) the system extracts

supported content; (3) teacher confirms class, subject, chapter and learning objectives; (4) AI proposes a structured lesson; (5) teacher previews and edits; (6) teacher reviews generated questions and explanations; (7) teacher publishes, or saves as draft. If extraction fails, the system explains the problem and preserves the original file.

Lesson editor capabilities (S5 §31): add/delete/reorder lesson sections; edit text; add examples and questions; attach approved resources; preview the student experience; save drafts; publish/unpublish per permissions; revise a lesson WITHOUT silently changing completed student records.

VS-63 Session Builder

CORE P1

Teacher selects: class and subject → chapter or concept → learning objectives → source material → release time (if required) → deadline → questions and completion criteria. The system can propose a lesson sequence; the teacher reviews and publishes. **Per-class question swap:** the STRUCTURE stays fixed, the QUESTIONS change per class — reuse across sections without duplication.

VS-64 Recording Upload & Segmentation

PROPOSED

Recording upload → cut into concept segments (manual or AI-assisted) → concept tags. Governed by the school-level permission setting (**VS-85**); the school decides whether recordings exist at all. Teachers should never be REQUIRED to record lessons (S5 §3).

VS-65 Homework & Quiz Creator

CORE P1

Create MCQ / short-answer / practice sets → deadline → assign → completion list generated automatically. Two question routes: the school question bank (**VS-87**) or AI drafts (**VS-43**) — both flow through teacher review. AI-generated questions are always reviewed before consequential use (S5 §34).

VS-66 Auto-Grading + Review Queue

PROPOSED

MCQs auto-grade instantly. Short answers enter a review queue where the AI optionally drafts feedback — *"The answer is in the right direction, the unit is missing"* — which the teacher edits before sending. **Payoff:** instead of 60 copies, the teacher handles only borderline cases. Teacher time returns. The review interface always distinguishes AI suggestions from teacher-approved grades (S5 §35: review submissions; inspect written answers; apply rubrics; edit AI-proposed feedback; override automated suggestions; publish final feedback; assign follow-up work).

VS-67 AI-Drafted Sessions & the School Knowledge Compiler

PROPOSED P2

The teacher uploads an EXISTING presentation → the AI drafts the session skeleton (slide breakdown, extracted definitions, drafted practice questions) → the teacher reviews and edits (about 20 minutes) → publish. Two hours of work becomes thirty. Old PPTs become living sessions — no new content or recording needed. **Non-negotiable:** the draft always passes through the teacher's hands — AI output never goes directly to a student.

The AI School Knowledge Compiler (S5 §32) is the expanded version: from one uploaded lesson, the AI may propose (1) a structured lesson, (2) a simplified explanation, (3) a detailed explanation, (4) a concept map, (5) prerequisite concepts, (6) practice questions, (7) misconception checks, (8) a short

revision sheet, (9) a guided self-study session, (10) a parent-friendly learning summary, (11) differentiated practice paths, (12) teacher review notes. Critical rule: generated material is traceable to its source; the teacher can review and reject unsupported content.

VS-68 Year-over-Year Content Reuse

PROPOSED

The "Reuse" button: dates update, questions swap, publish. First year is labour; from the second year it is an asset. Content compounds every year — after two years VidhyaSaathy becomes the workload-reducer it was not at the start. This compounding effect is the retention story.

VS-69 One-Tap Parent Updates

PROPOSED

SYSTEM DRAFT — TEACHER READS, EDITS, SENDS

"Aarav completed Session 3, quiz 4/5. Speed-velocity strong. Acceleration needs practice — a 20-minute revision is ready in the app."

ONE teacher, 60 parents, 10 minutes. **Tone design:** always constructive — "practice is assigned" — never "your child is behind" and never comparisons. The goal is parent SUPPORT, not a scolding excuse. (Blueprint S5 §73: a teacher seeing repeated difficulty may send a short update with one practical suggestion; access stays limited to each parent's own verified child.)

VS-70 Class-Level Mistake Autopsy & Analytics

PROPOSED ★

Autopsy (teacher side of VS-56): *"7 of 11 students have the SAME logic-break — they all interchanged velocity and speed. One misconception, one 5-minute clarification."* Matches against a misconception library — the documented misconceptions of physics/maths.

Class-level analytics (S5 §37): assignment completion; assessment distributions; concept-level performance; repeated errors; changes between assessments; participation in assigned activities; students who may benefit from extra support; outcomes after an intervention. Analytics support teaching decisions — never public rankings or conclusions about intelligence.

Targeted Reteaching Assistant (S5 §38): when the radar finds a shared gap, the AI drafts a short explanation, a worked example, a new diagnostic question, a 5-minute reteaching activity, extra practice, and a follow-up assessment. The teacher approves the plan and decides whether to use it.
Teacher Workload Assistant (S5 §39): drafting lesson structures from existing material; drafting feedback comments; generating question variations; summarising class-level gaps; preparing revision activities; creating parent-friendly progress summaries; organising resources by chapter and concept — reducing repetitive work while preserving teacher judgement and written practice.

VS-71 Teacher Mirror & Teacher DNA

PROPOSED

Teacher Mirror (teacher side of VS-55): the teacher's own class held up to them — *"You spent 12 minutes on acceleration, interaction 2 minutes. The quiz missed that concept the most. Connection?"* The teacher improves their own teaching from evidence — an insight even coaching institutes do not give their own teachers. **Teacher DNA** (VS-57, detailed in Section 6.8) surfaces to teachers as a profile they must approve before students see any priority hints derived from it.

VS-72 **Class Teacher Role & Announcements** CORE

Class teacher: extra rights for ONE class — broadcast class updates, manage the Class Space (**VS-10**), handle small disputes. Subject teachers work only inside their own subject workspaces. **Teacher announcements (S5 §40)** reach only the intended classes and users: upcoming sessions, assignment reminders, test dates, revision instructions, changes to learning resources. Assignment and session reminders go to students; parents receive selected reminders per their preferences and school policy (S5 §71).

7.15 Teacher Screen Connection Map

Blueprint teacher screens (S5 §88), in workflow order: Teacher home dashboard → Class list → Class detail → Student roster → Subject & chapter view → Resource library → Upload resource → AI-generated lesson preview → Lesson editor → Guided session builder → Session scheduler → Assignment builder → Question library → Assessment review → Student work review → Concept radar → Class analytics → Targeted reteaching assistant → Announcement composer → Teacher profile & preferences.

DAILY LOOP CONNECTION

Dashboard heat-map (red concept) → reteaching assistant (draft plan) → approve → assign follow-up quiz → next morning heat-map recheck. The insight-to-action loop is never more than 3 taps deep.

7.16 Teacher-World Checklist

- [ ] Dashboard leads with class-level gaps, not vanity metrics
- [ ] Publishing a note takes the tagged few-tap flow (VS-62)
- [ ] Session builder enforces the 10-step structure + release time
- [ ] Per-class question swap preserves structure
- [ ] Short-answer review queue separates AI drafts from teacher grades
- [ ] AI session skeletons always require explicit teacher approval
- [ ] “Reuse” button reproduces last year’s session with new dates
- [ ] Parent update drafts are constructive-only by template design
- [ ] Class insights anonymous by default (pending the OPEN decision)
- [ ] Mid-year teacher reassignment handled without data loss (VS-82)

PART 8 · PARENT

Parent World

Parents know what the child is studying, so the child is not lost in “YouTube chaos” at home. Parents do not become the child’s teacher — their role is SUPPORT, not replacing school teaching.

VS-73 – VS-78

VS-73 Parent Dashboard — plain language by design

CORE

What parents SEE

What the child is studying (subject + chapter) · what is due, and what is completed · quiz results in simple framing · where extra practice is needed · teacher updates for the class · actionable "do this" suggestions.

What parents NEVER see

Anything about other children · leaderboards or rankings · what the child discussed with the AI — private by design (**VS-78**).

Design insight: the parent is not a data expert. "Concept mastery 63%" means nothing to them. What must appear is: *"Biology strong, one chapter weak in Physics — revision is ready in the app."* DATA MUST BECOME A DECISION, NOT A DISPLAY. The Blueprint parent dashboard (S5 §41) answers five questions — What is my child currently learning? What has been completed? Where might they need help? What is the teacher doing about it? How can I help without adding pressure?

VS-74 Weekly Progress Summary & Parent Reports

PROPOSED

Weekly digest: what was completed, what improved, what is coming next week. Frequency (daily / weekly / on-event) is an OPEN team decision — Section 17.4.

PARENT PROGRESS REPORT SAMPLE (S5 §42)

"Your child can recall the definitions of speed and velocity but needs more practice applying them to word problems. The teacher has assigned a short practice session for this concept."

The report distinguishes: directly observed results / AI-generated interpretations / teacher-confirmed conclusions / suggested actions — and never presents an inference as a medical, psychological or permanent ability diagnosis. **Parent Translator (S5 §63)**, the engine behind these reports: explains what the evidence shows; identifies uncertainty; mentions the teacher's plan; suggests ONE realistic support action; avoids alarming language and labels.

VS-75 "Ask This" Parent Prompts

PROPOSED

ONE SIMPLE DAILY/WEEKLY SUGGESTION

"Today, ask Aarav: what is the difference between acceleration and velocity? Whoever explains it, remembers it better."

The quiet genius: what you teach someone else is your most durable learning (the protege effect). The parent does not need to know the subject — only how to ask the question. Blueprint S5 §43 adds the recommendation menu: ask the child to explain one concept in their own words; review an assigned example together; encourage completing a teacher-assigned session; contact the teacher if a difficulty persists; recognise improvement rather than comparing marks.

VS-76 **WhatsApp/SMS Fallback** PROPOSED P3

NO APP? KEY UPDATES STILL ARRIVE

"Aarav — Science homework pending (last day tomorrow). Quiz 4/5. Today's revision: Acceleration, 20 minutes."

Why India-critical: not every parent is app-savvy, especially in Tier-2/3 cities. WhatsApp Business API is ubiquitous in India. Parent engagement has a ceiling without this — roughly 30% of parents will never open the app; for them, the fallback decides whether the parent layer exists at all.

VS-77 **PTM Time Machine — Parent Mode** PROPOSED

The parent side of **VS-59** (Section 6.8): simulated PTM practice with three trained, data-backed questions, so the real 6-minute meeting is confident and specific instead of awkward and generic. The student mode lets the child rehearse an evidence-based growth story.

VS-78 **Parent Privacy Boundaries — hard rules** CORE

- Parent-child AI conversations are NEVER automatically visible — not even through a setting. By design.
- Parents see ONLY their own verified child.
- Private student-teacher notes are never shown to parents.
- Sensitive learning details appear only through transparent school policy, age-appropriate safeguards and applicable legal requirements (S5 §45).
- These rules are not marketing — they are baked into the architecture (RBAC, **VS-07**; Part 13).

Parent notifications (S5 §44) — allowed types: new assignment; upcoming deadline; important teacher announcement; published progress report; significant learning milestone; teacher-requested follow-up. Preferences and school rules are configurable; the app avoids alerting on every click, question or minor score change.

8.7 Parent Screen Connection Map

Blueprint parent screens (S5 §89): Parent home dashboard → Linked-child view → Learning progress → Assignment list → Assessment history → Parent-friendly learning summary → Teacher-approved progress notes → Notification settings → Account & relationship settings. Entry: OTP verification (**VS-05**) or WhatsApp fallback (**VS-76**) for non-app-users.

8.8 Parent-World Checklist

- [ ] Every parent-facing sentence passes the "would a non-expert parent act on this?" test
- [ ] Zero access to other children's data, rankings, or AI chats
- [ ] Weekly digest template + frequency decision recorded
- [ ] "Ask this" prompt engine fed by real learning evidence
- [ ] WhatsApp fallback delivers the 3 core message types
- [ ] PTM practice mode reads from the child's actual data
- [ ] Notification defaults calm (no per-click alerts)

PART 9 · ADMIN

School Admin World

Structure, not surveillance. The admin sees the school's organisation and health — never student marks, never teacher league tables.

VS-79 – VS-87

VS-79 Admin Dashboard

CORE P1

The school's structure in one place: classes, teachers, enrolment statistics, parent-linking status, resource organisation. NO student ranking, NO teacher surveillance — structure and health only. Blueprint additions (S5 §17): configured classes and sections; active teachers and enrolled students; account invitation status; subjects and teaching assignments; published learning resources; general platform activity and operational issues; basic adoption indicators.

VS-80 Academic Year & Calendar Management

CORE

A new year = roll over the old structure + new students + new dates. This is where **VS-92** (Calendar Awareness) is fed from: exams, holidays, festivals. Academic year configuration is one of the 15 school-profile capabilities in S5 §14 (see 9.7).

VS-81 Class & Section Creation

CORE

9-A, 9-B, sections, streams — a mirror of the school's real structure. Wizard work measured in minutes. Grade/class and section creation, subject creation and roster management are all admin functions (S5 §14).

VS-82 Teacher Assignment

CORE

Teacher × subject × class mapping. Changing it must be EASY — mid-year teacher changes happen constantly in India; the app has to handle them without breaking workspaces, sessions or histories.

VS-83 Student Enrolment — CSV + Activation Codes

CORE P1

How: CSV upload → validation (duplicates, missing parents, malformed rows) → activation codes generated → distribution sheet downloads → school distributes codes → parents verify by OTP → accounts live. Mid-year admission takes about 2 minutes end-to-end. **CSV columns:** name, roll number, class, parent phone number (**VS-01**).

Open decision: who adds students — admin upload, class teacher, or student self-join with a code? All three sources leave this undecided (Section 17.4). The Blueprint's bulk import (S5 §15) specifies: import from a validated spreadsheet; import teacher and class assignments; detect missing required fields; detect duplicate or conflicting records; preview changes before confirming; report failed rows with understandable reasons; generate school-approved invitations or credentials; allow authorised corrections after import. Bulk import is a scale-up feature; the first prototype ships with seeded users.

VS-84 **Parent Linking Management** CORE

Tracks: which parent is linked to which child, phone verification status, two-parents-one-child cases, and GUARDIAN cases (uncle, grandfather — common in India; a reality generic products miss). Parent-student relationship management is an admin capability (S5 §14).

VS-85 **Permissions & Access Control** CORE

The ROLE MATRIX: who can see what. School-level settings include: Are classroom recordings allowed at all? (VS-12) · Doubt clustering anonymous or named? (VS-37) · Parent update frequency (VS-74) · Resource and lesson ownership; publication and approval statuses; school-level settings; role and permission management; account suspension or deactivation where authorised (S5 §14). **Principle:** the school gets control; the platform provides guardrails.

VS-86 **School-Level Academic Reports** PROPOSED

Completion trends, subject health, resource coverage — FOR improving learning and supporting teachers, never for ranking or punishing students or staff. School reports mirror the platform's "structure, not surveillance" stance.

VS-87 **The School's Own Question Bank** PROPOSED ★

Every quiz, homework and approved AI draft (VS-43) feeds the bank: **Year 1: ~200 questions. Year 2: ~500** — all in the school's syllabus, notation and teacher style. After three years: the school's most valuable digital asset.

STRATEGIC VALUE

Features can be copied — this bank cannot. Every year of use makes leaving VidhyaSaathy harder. The realistic path to becoming defensible.

9.10 Admin Screen Connection Map

Blueprint admin screens (S5 §90): School dashboard → School profile → Academic year settings → Class & section management → Subject management → Teacher management → Student management → Parent linking → Bulk import → Invitation status → Resource permissions → School settings → Audit / administrative activity view (where appropriate).

9.11 Admin-World Checklist

- [ ] New-year rollover preserves histories and re-links assignments
- [ ] CSV validation reports every failed row with a human reason
- [ ] Activation codes are one-time use and expiring
- [ ] Guardian relationships modelled beyond mother/father
- [ ] Role matrix editable per school; defaults safe
- [ ] Recording permission default follows school decision, not platform
- [ ] Reports contain zero student rankings and zero teacher league tables
- [ ] Admin cannot view student marks or AI conversations anywhere

PART 10 · INDIA LAYER

The India Layer

These features exist because the product is built FOR India, not "about" India from research. Design must consider uneven internet quality, shared family phones, mobile-first use, teachers with little time, multiple languages, varied digital literacy and different school budgets.

VS-88 – VS-92

VS-88 Offline Mode + Smart Sync

PROPOSED · P3 · ★

Mechanics: at school the internet is good → the student presses "Download today's session" → text-first + compressed images + optional low-res video → about 2–3 MB → at home with no internet, the cached session runs → the quiz is attempted OFFLINE → answers saved locally → internet returns → SYNC → data reaches the teacher dashboard. **Design rules:** the download button is EXPLICIT — the child decides when to spend data. Conflict rules are simple: timestamps, last-write wins. **Why non-negotiable:** in India this is not a feature — it is an ACCESS requirement. (The Blueprint phrases offline as a possible future capability for the first pilot — S5 §84 — while the Master Spec schedules it in Phase 3; the team must decide if the pilot needs it — Section 17.4.)

VS-89 Data Saver + Audio-Only Mode

PROPOSED

Every resource shows its size ("Session 4: 2 MB"). Recordings have an AUDIO-ONLY mode — on the bus, out of data? Listen to the recording like a podcast. 90% of an explanation video's value is in its audio. Small feature, adoption-level difference.

VS-90 Language Layers — three separate things

PROPOSED

1. **UI LAYER** — the interface itself in Hindi/regional languages.
2. **AI LAYER** — ask in any language, answered in that language (VS-35).
3. **BRIDGE LAYER** (VS-47) — English notes stay English (they are board English); the Hinglish/regional EXPLANATION toggle is the real differentiator. "Textbook in English, understanding in your language."

Layers 1–2 anyone can build. Layer 3 — grounded in the teacher's own material — nobody else is doing.

VS-91 Low-End Device Optimization

PROPOSED

PWA or lightweight app. Smooth on a 2GB-RAM budget Android. No heavy animations, WebP images, and a hard target: **EVERY SCREEN LOADS IN 2 SECONDS.** The phone a student studies on already runs 15 other apps.

VS-92 **Academic + Festival Calendar Awareness** PROPOSED

The system KNOWS today is Diwali, or that it is exam week. Session deadlines never land on holidays; streaks freeze automatically during exam week (**VS-93**); "next class" calculations are calendar-aware. A small feature that signals whether the product was built for India or merely about it.

10.6 Accessibility & Connectivity Requirements (S5 §84)

Usable on low-bandwidth connections and ordinary devices; readable typography; keyboard and screen-reader support where feasible; clear focus states; captions or transcripts for audio/video where feasible; no reliance on colour alone to convey state. The India layer checklist:

- [ ] Offline session download + sync works end-to-end on a real low-end phone
- [ ] Audio-only mode plays recordings as podcasts
- [ ] UI language + AI language + explanation bridge independently set
- [ ] App usable on 2GB-RAM Android; screens ≤ 2s on 3G-class networks
- [ ] Calendar blocks deadlines on holidays; freezes streaks in exam week

PART 11 · MOTIVATION

Motivation, Games and Learning Design

Growth, not rank. The goal is habit-building, not addiction — no public leaderboards, no shame-based competition, anywhere in the product.

VS-93 – VS-100

VS-93 **Streaks with Freeze** PROPOSED P4

Daily study streaks — but automatically FROZEN during illness, exam week and holidays (fed by **VS-92**). Returning resumes the streak where it was. **Why:** normal streak design punishes — one missed day resets everything. In India, load-shedding, shared devices and family events are NORMAL. The goal is habit-building, not addiction.

VS-94 **Improvement Badges — growth, not rank** PROPOSED

NO	YES
"Topper badge — 95%" (the already-strong child wins every time)	"Acceleration: 30% → 65% — GROWTH BADGE!" (goes to whoever needed it most)

A child who climbed from weak to average worked harder than the topper — without reward, they quit the app. No public leaderboards, no unhealthy competition — a rule running through every source document. Blueprint achievements (S5 §68): completing a learning recovery path; demonstrating a previously difficult concept; improving an explanation; correcting a repeated error; finishing an assigned chapter activity. **Never reward time spent, click counts or raw speed.**

VS-95 **Daily 5-Minute Challenge** PROPOSED

Every day, five questions: 2 from today's topic + 1 revision (earlier material) + 1 prerequisite check + 1 bonus. The daily loop stays alive without overload — spaced repetition, simply baked in. A child who gives even 5 minutes daily stays in the chapter's flow all week.

11.4 VS-96 to VS-100 — The Five Games (One Question Bank)

Games demand NO separate content — they run on the teacher's approved questions (**VS-87**); only the wrapper changes:

ID	Game	What it does	Helps
VS-96	**MCQ Rush**	Rapid questions against a timer, streaks.	Speed + recall
VS-97	**Match-the-Concept**	Match terms to formulas/definitions.	Vocabulary, formulas
VS-98	**Rapid Revision**	10 mixed questions from the chapter.	Pre-test revision
VS-99	**Streak Challenge**	One wrong answer resets the streak.	Consistency
VS-100	**Boss Level**	All concepts mixed; pass = next chapter unlocks.	Sequential mastery

THE RULE

Games SUPPORT lessons — they never become the product. Points reward meaningful practice, never screen time. After every game: *"Today you answered 15 Acceleration questions — 12 correct. Tomorrow's target: the Match game."*

The Team Guide's earlier illustrative game set maps onto these five formats — the list was always illustrative, 4–5 games that shift slightly per chapter.

11.5 Activities, Adaptive Practice, Hints, Reflection

- **Chapter activity system (S5 §65):** reusable activity types — multiple-choice challenges; matching concepts; order-the-steps; identify-the-error; prediction questions; scenario-based problems; short explanation prompts. Activities are chosen by learning objective — never to make the interface look entertaining.
- **Adaptive practice (S5 §66):** see **VS-23** / **VS-46** ; the action menu includes another example, changed representation, prerequisite return, harder problem, hint, or teacher support, with recommendations explained.
- **Hints and scaffolded help (S5 §67):** the hint ladder reveals enough to move forward without giving away the answer: (1) point to the relevant concept; (2) ask a guiding question; (3) show the first step; (4) offer a worked example; (5) reveal a full explanation when appropriate. Teachers set different help rules for practice vs formal assessments.
- **Reflection (S5 §69):** after selected activities — What did you understand? What is still confusing? Which step helped? Can you explain it in your own words? Reflections inform the next step — but self-reported confidence is never treated as proof of mastery.

11.6 Motivation Checklist

- [ ] Streak freezes on illness / exam-week / holiday automatically
- [ ] Badge engine triggers on DELTA, not absolute score
- [ ] Zero public leaderboards anywhere in the product
- [ ] All five games pull only approved questions
- [ ] Hint ladder configurable per assessment type
- [ ] Reflection answers feed next-step logic, never mastery flags

PART 12 · FLOWS

End-to-End Flows: The Whole System in Motion

All students, scores and times are fictional. Consolidated from the Master Spec (S4 Part 10) and the Team Guide's "A day in Class 9-A".

12.1 A Day in Class 9-A, Minute by Minute

07:00 ADMIN	A new mid-year admission arrives: add to CSV → activation code → parent OTP → account live. Two minutes of work. VS-83 · VS-04 · VS-05
10:00 SCHOOL · PHYSICS PERIOD	Sharma Ma'am teaches Motion — Distance, Speed-Velocity, Acceleration. The class is recorded (school permitting). VS-12
14:00 TEACHER · STAFF ROOM	Ma'am opens the app: uploads the recording (AI suggests segments — "Acceleration: 12:40–19:30") → publishes concept-tagged notes → creates SESSION 3 from an AI-drafted skeleton based on her old PPT, edited in 20 minutes → "Opens: 6 PM tonight, due: tomorrow morning" → publish → class notification. VS-64 · VS-62 · VS-67 · VS-63 · VS-16
18:00 THE SESSION OPENS	The time-controlled release rule fires; Session 3 becomes available to the class. VS-16 · Section 5.4
18:30 STUDENT A — STRONG	Dashboard next action: "Session 3 — 15 minutes." Slides → in-session questions → QUIZ 5/5 → Confidence Meter: was "Sure", and correct → the system recommends a HARDER Speed-Velocity problem set (L3) and unlocks Boss Level. VS-15 · VS-39 · VS-23 · VS-100
18:30 STUDENT B — STRUGGLING	QUIZ 2/5 — Acceleration wrong. Opens the "Why was it wrong?" feedback → re-reads the explanation slide → GAP DETECTOR checks: the velocity questions WERE answered correctly → the problem is Acceleration itself → recommendation: simpler explanation + 3 worked examples + targeted practice. Plus MISTAKE DNA: "2 concept mistakes, 1 careless (sign)" — separate fix routes. VS-18 · VS-30 · VS-40
18:30 STUDENT C — WAS ABSENT	Top dashboard card: "You weren't in class yesterday. Catch up in 20 minutes?" → the 7-minute Acceleration recording segment → notes → mini session → 5-question check → pass → back at class pace. VS-20 · VS-12 · VS-11 · VS-16

21:00 NIGHT RECAP

All three get personalised audio: A gets tomorrow's preview; B gets the fix for today's acceleration mistake; C gets "catch-up complete, ready for tomorrow's class." VS-49

21:30 PARENT OF STUDENT B · WHATSAPP

"Session 3 complete, quiz 2/5. Acceleration practice is assigned — 20 minutes, ready in the app. Today, ask: what's the difference between acceleration and velocity?" — plain language, with a next step. VS-76 · VS-75

07:30 NEXT MORNING · TEACHER

Heat-map: ACCELERATION IS RED — 60% of the class wrong. Doubt clustering: 17 students asked Acceleration doubts overnight, anonymously. Mistake autopsy: 7 of 11 share the same confusion — velocity/speed interchange. Ma'am reteaches Acceleration for 10 minutes and clarifies exactly that misconception → retest → heat-map turns green. **THE LOOP IS COMPLETE** — and the next class is built on this evidence. This is the whole app: after the bell rings, the school speaks again. VS-61 · VS-37 · VS-70

12.2 The Motion Chapter Worked Example (13 Steps, S5 §80)

Class 9, Physics, Motion; concepts: distance & displacement, speed & velocity, acceleration. Fictional Indian school.

Step	Who	What happens in VidhyaSaathy
1	Teacher	Organises the Motion lesson material; links a recording and a chosen video to the chapter.
2	AI	Prepares a draft package: structured lesson, definitions and examples, concept map, practice questions, diagnostic check, guided self-study session.
3	Teacher	Reviews and corrects the material, approves the questions, schedules the session.
4	Students	Receive the session on their dashboards.
5	Student	A completes the diagnostic with strong results; receives a more challenging practice activity.
6	Student	B confuses speed and velocity in a scenario question.
7	AI	Links the answer to the concept; offers a different explanation with an everyday example.
8	Student	B tries again on a new question testing the same distinction in a different context.
9	System	Updates its estimate: the new response is additional evidence about understanding.
10	Teacher	Sees class-level patterns — is the confusion isolated or common?
11	Teacher	Publishes a short reteaching activity if several students share the difficulty.
12	Parent	Sees the assigned activity and progress summary — never the private AI chat transcript.
13	ALL	A later assessment checks whether the concept is now understood. **THE LOOP REPEATS.**

This is the central VidhyaSaathy experience: one classroom lesson leading to different, evidence-based learning paths, with the teacher and parent appropriately informed. (The Vision document's variant adds that the next lesson begins with a brief acceleration recap built on this evidence.)

12.3 The Loop in Practice — Design Rule

LEARN — the teacher's lesson, notes, recording or chosen resource

↓

PRACTISE — questions and worked examples

↓

UNDERSTAND — assessment shows which concepts are secure

↓

PROGRESS — the next step: revision, prerequisite work, more practice, or a challenge

Assessment exists to decide WHAT HAPPENS NEXT, not only to assign marks. Progress is tracked per concept, not only per test.

12.4 Flow Checklist

- [ ] The full-day demo (12.1) runs end-to-end without manual data patching
- [ ] Each of the three student archetypes reaches a distinct next step
- [ ] Teacher's morning dashboard reflects the previous night's evidence
- [ ] Parent message generated automatically, constructive tone
- [ ] Reteach → retest → heat-map colour change demonstrable in one cycle

PART 13 · SECURITY

Data, Security and Responsible AI

Privacy is a product requirement, not a patch. Permissions are enforced at the data and server layers — a client-side role label is never proof of anything.

13.1 Roles and Permissions Enforcement (S5 §81)

- Students access their own learning records and authorised class resources only.
- Teachers access assigned classes and authorised student work only.
- Parents access only correctly linked children and permitted progress information.
- School administrators manage their school's authorised users and structure.
- Platform administrators receive only the access required for legitimate platform operations.

> **IRON RULE**
> Changing a client-side role label must NEVER grant additional access.

13.2 Student Data Protection (S5 §82)

The system should: collect only necessary information; protect account credentials; restrict access to student records; use secure transport and appropriately protected storage; keep audit records for sensitive administrative actions where appropriate; provide processes for account correction and

deletion; define retention rules; protect backups and exported records; and avoid exposing sensitive data in logs or error messages.

The implementation must follow applicable Indian privacy and child-data requirements, with appropriate school and guardian processes. Privacy principles from the vision document: verified student and parent access; role-based permissions; protection of records; minimal data collection; teacher oversight of important educational decisions; transparent use of learning data; responsible handling of recordings; age-appropriate design. **The platform must never use invasive monitoring, public rankings of struggling students, or unrestricted sharing of academic information.**

13.3 AI Safety and Reliability (S5 §83)

Consolidated with **VS-32** in Section 6.10 — the full rule list: grounding in approved sources; visible uncertainty; no fabricated citations; AI suggestions separated from teacher-approved content; no medical/psychological diagnoses; no permanent labels; age-appropriate explanations; student data protected from unnecessary disclosure; prompt injection in uploaded documents treated as untrusted input; graceful fallback when the AI service is unavailable.

13.4 Evidence and Analytics Integrity (S5 §85)

FIVE STATES THAT ARE NOT INTERCHANGEABLE

1 · Activity completed 2 · Question answered correctly 3 · Concept demonstrated 4 · Teacher-verified mastery 5 · AI-inferred possible difficulty.

The app must never manufacture precision or present a weak inference as a fact. This rule is why video-watching, slide-opening and session completion are recorded separately from mastery throughout Part 5.

PART 14 · ARCHITECTURE

Technical Architecture

A pragmatic starting stack, the data model the product implies, and the failure behaviour every screen must honour.

14.1 Recommended Starting Stack (S5 §91 + Team Guide S1 §13)

Layer	Blueprint recommendation
Frontend / app	TypeScript; React or Next.js; responsive web; Tailwind CSS or equivalent styling.
Backend / data	Supabase with PostgreSQL: authentication, database access policies, file storage for approved uploads, server-side functions for privileged operations.
AI layer	Server-side AI integration; retrieval of relevant approved school content; structured outputs where useful; source tracking and validation; deterministic calculations for simulations; fallback responses for failed API calls.
Deployment	Suitable web hosting; environment variables for credentials; separate dev/prod configuration; error logging that avoids unnecessary student data.

IRON RULE

API keys must NEVER be placed in browser-visible code; use only services actually configured by the team.

Team Guide options (equally valid, pick what the team knows best): frontend React/Next.js or Flutter (mobile-first matters for students); backend Node.js or Django; database MongoDB or PostgreSQL (relational fits classes, sessions and progress); AI via OpenAI-style API with custom context. AR/VR classrooms remain cut (Part 15.3).

14.2 Core Data Entities (S5 §92) — 37

People & structure (1–14)

1 Users · 2 Schools · 3 School memberships · 4 Student profiles · 5 Teacher profiles · 6 Parent-student relations · 7 Academic years · 8 Classes · 9 Sections · 10 Subjects · 11 Teacher assignments · 12 Chapters · 13 Concepts · 14 Concept prerequisites

Content & learning (15–24)

15 Resources · 16 Resource versions · 17 Lessons · 18 Guided sessions · 19 Session sections · 20 Session enrolments/assignments · 21 Assignments · 22 Questions · 23 Question options · 24 Assessment attempts

Evidence & communication (25–34)

25 Student responses · 26 Feedback · 27 Concept evidence · 28 Learning plans · 29 Learning-plan steps · 30 Student progress events · 31 Teacher interventions · 32 Announcements · 33 Notifications · 34 Parent progress summaries

AI & audit (35–37)

35 AI interaction metadata (per privacy policy) · 36 Learning Passport entries · 37 Administrative audit events

Not every entity must exist in the first prototype; design the production schema carefully before adding complex analytics.

14.3 AI Request Workflow

See Section 6.10 — the 11-step server-side sequence from authentication through source-attachment to outcome recording.

14.4 Failure and Fallback Behaviour (S5 §94)

The application must handle: AI service unavailable; invalid AI response; failed file upload; unsupported document format; interrupted network request; expired session; permission denied; missing school content; failed assessment submission; database or storage error. **In every case the user receives an understandable message and a safe next step; unsent work is preserved or a retry is offered where appropriate.**

PART 15 · BUILD ORDER

Build Order, Phases and Deliberate Cuts

The loop must exist first; delight comes after. Scope discipline is what saves the product.

15.1 Master Spec Build Order (S4 Part 12)

1. **PHASE 1 (RED) — CORE LOOP.** Without it the product does not exist: VS-01 to VS-09, VS-16, VS-17, VS-18, VS-27, VS-60/VS-61. One school, one chapter, one session, one quiz, one teacher insight — the whole loop runs.
2. **PHASE 2 (ORANGE) — AI BRAIN** (differentiation + pitch): VS-28/29/30 (context AI, Socratic, gap detector), VS-31, VS-37, VS-39 (Confidence Meter), VS-44 (Interest Remix), VS-36 (SnapDoubt), VS-40 (Mistake DNA). Demo story: "we don't just grade — we understand the STUDENT'S THINKING."
3. **PHASE 3 (YELLOW) — INDIA LAYER** (adoption): VS-88 (offline), VS-89 (data saver), VS-47 (Hinglish bridge), VS-76 (WhatsApp), VS-06 (shared profiles). The layer that scales.
4. **PHASE 4 (GREEN) — ENGAGEMENT + SIGNATURE:** Streaks, badges, games, Recap, Viva, Digital Twin, Reverse Class, PTM Time Machine.

15.2 Blueprint Build Priorities (S5 Part 16)

- **Phase 1 — six-hour hackathon prototype:** one complete, convincing learning loop, not many disconnected screens — polished landing/sign-in; seeded demo school and Class 9-A; three role dashboards; a published Motion lesson; one guided session; a short diagnostic quiz; two seeded student profiles with different results; a personalised explanation or recovery path; a simple concept-level teacher dashboard; a parent-friendly progress summary; a working database connection where feasible; a deterministic AI fallback; a polished cloud mascot and consistent theme; a tested end-to-end demo. Sample student data must be clearly labelled as simulated — no claims of improved real outcomes.
- **Phase 2 — functional school pilot:** real authentication; school/class setup; secure teacher/student permissions; resource uploads; lesson creation and publishing; guided session scheduling; assignments and assessments; persisted responses; basic parent-student linking; teacher-reviewed AI content; reliable progress tracking; basic admin tools.
- **Phase 3 — intelligent learning system:** concept/prerequisite graph; learning-state model; misconception detection; Learning Recovery Engine; adaptive practice; AI-generated lesson packages; classroom concept radar; reteaching assistant; evidence-based parent summaries; Learning Passport.
- **Phase 4 — signature AI experiences:** What-If Machine; Mistake Time Machine with supported handwriting analysis; Chapter-to-Comic Studio; Infinite Textbook Adventure; Concept Court; Reverse Exam Mode; Viva Simulator; Reality Scanner; Learning Remix; AI Debate. Prioritise by user testing, educational value, cost and reliability — not impressiveness.
- **Phase 5 — scale and school adoption:** bulk onboarding; more grades and curricula; additional Indian languages; improved low-bandwidth behaviour; optional offline learning and sync; school-to-school portability of permitted records; more sophisticated analytics; practical integrations with existing school systems.

15.3 Features Deliberately Cut — the Discipline List (S4 Part 11)

Cut	Why
Coaching-institute integration	Directly clashes with the mission: coaching dependence should DECREASE, not be integrated.

Camera/mic emotion detection	Privacy nightmare with children; the behaviour-based VS-33 is its safe version.
Public class leaderboards	The documents' own rule: no unhealthy competition. Improvement badges (VS-94) replace them.
Transport / fees / ERP modules	The documents are explicit: this is not an ERP. Scope discipline saves the product.
AR/VR, study groups, peer teaching	Good ideas, outside the MVP — Phase 3+ material.

15.4 The One-Line Truth (S4 Closing)

> *101 features, one single thread: the teacher's effort is applied once → the system reuses it again and again → the AI adds the school's context → evidence decides the next step → and that 10-minute reteach Ma'am does tomorrow morning is driven by data the night itself generated.*

PART 16 · QA

Acceptance Criteria and QA Checklists

The master checklists from the Blueprint (S5 Part 17) — the definition of "working" for this product.

16.1 Account and Permission Tests

- [ ] A student cannot open teacher-only functions by changing the URL.
- [ ] A parent can see only the child or children correctly linked.
- [ ] A teacher can access only authorised classes and student work.
- [ ] Signing out prevents access to protected pages.
- [ ] Failed authentication does not reveal sensitive account information.

16.2 Learning Workflow Tests

- [ ] A teacher can publish a lesson to the intended class.
- [ ] An authorised student can open the lesson.
- [ ] The student can submit a supported answer.
- [ ] The system records the answer and its result.
- [ ] The student receives the appropriate next step.
- [ ] The teacher can view authorised class-level results.
- [ ] The parent sees the permitted progress summary.
- [ ] The app distinguishes activity completion from demonstrated mastery.

16.3 AI Reliability Tests

- [ ] AI explanations use relevant approved content where available.
- [ ] Unsupported questions never trigger fabricated source claims.
- [ ] AI-generated lesson drafts are reviewable before publication.

- [ ] Numerical simulations use deterministic calculations.
- [ ] Uncertain handwriting/solution parsing is confirmed by the student.
- [ ] The system handles AI failure without losing completed work.

16.4 User Experience Tests

- [ ] The primary student action is easy to find.
- [ ] The teacher can publish a simple session without unnecessary steps.
- [ ] The parent understands the progress report.
- [ ] The interface works on common phone and laptop screen sizes.
- [ ] Text remains readable and actions have clear feedback.
- [ ] The cloud companion does not obstruct learning.
- [ ] Errors explain what happened and what the user can do next.

16.5 Product Validation Tests — before claiming improved learning

- [ ] Can students complete a guided session independently?
- [ ] Does a personalised explanation improve performance on a fresh question?
- [ ] Do teachers save time compared with their existing workflow?
- [ ] Do teachers trust the generated content after review?
- [ ] Do parents understand the progress summaries?
- [ ] Does the product work reliably on the devices and connections students actually use?
- [ ] Does the learning loop produce better outcomes than simply sharing a PDF or video?

PART 17 · REGISTRY

Feature Registry and Cross-References

The complete VS-01–VS-100 index, the role × feature matrix, the Blueprint-only modules, the open decisions register, and the six key workflow chains.

17.1 Complete VS Feature Index (VS-01 to VS-100)

Tags: **C** = Core (founder-confirmed) · **P** = Proposed · build phase per the Master Spec order · ★ = source emphasis (★ highlighted, ★★ flagship, ★★★ highest).

ID	Feature name	Part	Tag
VS-01	School Onboarding Wizard	3.2	C · P1
VS-02	School Admin Account & Login	3.3	C · P1
VS-03	Teacher Account & Login	3.4	C · P1
VS-04	Student Verification & Login ★	3.5	C · P1
VS-05	Parent Account Linking	3.6	C · P1
VS-06	PIN Shared-Device Profiles	3.7	P · P3
VS-07	Role-Based Access Control	3.8	C · P1

VS-08	Board-Wise Curriculum Mapping	3.9	C · P1
VS-09	Structured Class Workspace	4.1	C · P1
VS-10	Class Space (Same-Class Visibility)	4.2	C
VS-11	Concept-Tagged Teacher Notes	4.3	C · P1
VS-12	Segmented Classroom Recordings	4.4	P
VS-13	Trimmed Curated YouTube	4.5	P
VS-14	Persistent Chapter History	4.6	C
VS-15	Student Dashboard	5.1	C · P1
VS-16	Guided Self-Study Sessions ★★	5.3	C · P1
VS-17	In-App Homework	5.5	C · P1
VS-18	Post-Session Quiz + Why-Wrong Feedback	5.6	C · P1
VS-19	Two-Line Summarizer	5.7	P
VS-20	Missed-Class Recovery Path ★	5.8	C · P2
VS-21	Visual Concept Map	5.9	P
VS-22	Four Performance Bands (internal)	5.10	P
VS-23	Adaptive Difficulty Engine	5.11	P
VS-24	Exam-Format Practice	5.12	P
VS-25	Test Readiness Signal	5.13	P
VS-26	Session Completion Rules	5.14	C · P1
VS-27	Student Progress View	5.15	C · P1
VS-28	Context-Grounded Ask-Me-Anything ★	6.2	C · P2
VS-29	Socratic Mode ★	6.2	C · P2
VS-30	Prerequisite Gap Detector ★★	6.2	C · P2
VS-31	Study Plan Generator	6.2	C · P2
VS-32	AI Honesty Layer	6.1	C
VS-33	Frustration-Aware AI	6.2	P · P2
VS-34	Warmup Radar (Forgetting Detector)	6.2	P
VS-35	Voice Doubt Asking	6.3	P · P2
VS-36	SnapDoubt (Ask by Photo)	6.3	P · P2
VS-37	Doubt Clustering ★	6.3	P · P2
VS-38	Doubt Inbox (Opt-In Named)	6.3	P
VS-39	Confidence Meter (“Pakka Meter”) ★★	6.4	P · P2

VS-40	Mistake DNA (“Galti DNA”) ★	6.4	P · P2
VS-41	Rote-Learning Detector (“Ratta”) ★	6.4	P
VS-42	Avoidance Radar (“Darr Ka Map”) ★★	6.4	P
VS-43	AI Question Generation (Drafts)	6.5	P · P2
VS-44	Interest Remix ★	6.5	P · P2
VS-45	Personal Enemy Questions (“Dushman”) ★	6.5	P
VS-46	Adaptive Difficulty (Practice link)	6.5	P
VS-47	Hinglish Bridge ★	6.6	P · P3
VS-48	“Show You Understand” (Feynman) ★	6.6	P
VS-49	Night Recap (5-Min Podcast)	6.6	P · P4
VS-50	AI Big Brother/Sister (“Bhaiya-Didi”) ★★	6.6	P
VS-51	At-Home Viva (AI Oral Exam) ★	6.7	P · P4
VS-52	Exam Hall Simulator ★	6.7	P · P4
VS-53	Reverse Class (AI as Weak Student) ★★	6.7	P
VS-54	Digital Twin (Tomorrow’s Result) ★★	6.7	P · P4
VS-55	“Teacher Said So” Emphasis Radar ★★	6.8	P
VS-56	Mistake Autopsy (Thinking Path) ★	6.8	P
VS-57	Teacher DNA ★★	6.8	P
VS-58	Syllabus GPS ★	6.8	P
VS-59	PTM Time Machine ★★★	6.8	P
VS-60	Teacher Dashboard	7.1	C · P1
VS-61	Concept Heat-Map	7.2	C · P1
VS-62	Publishing Notes & Resources	7.3	C · P1
VS-63	Session Builder	7.5	C · P1
VS-64	Recording Upload & Segmentation	7.6	P
VS-65	Homework & Quiz Creator	7.7	C · P1
VS-66	Auto-Grading + Review Queue	7.8	P
VS-67	AI-Drafted Sessions	7.9	P · P2
VS-68	Year-over-Year Reuse	7.10	P
VS-69	One-Tap Parent Updates	7.11	P
VS-70	Class-Level Mistake Autopsy ★	7.12	P
VS-71	Teacher Mirror	7.13	P

VS-72	Class Teacher Role	7.14	C
VS-73	Parent Dashboard	8.1	C
VS-74	Weekly Progress Summary	8.2	P
VS-75	“Ask This” Parent Prompts	8.3	P
VS-76	WhatsApp/SMS Fallback	8.4	P · P3
VS-77	PTM Time Machine (Parent Mode)	8.5	P
VS-78	Parent Privacy Boundaries	8.6	C
VS-79	Admin Dashboard	9.1	C · P1
VS-80	Academic Year & Calendar	9.2	C
VS-81	Class & Section Creation	9.3	C
VS-82	Teacher Assignment	9.4	C
VS-83	Student Enrolment (CSV + Codes)	9.5	C · P1
VS-84	Parent Linking Management	9.6	C
VS-85	Permissions & Access Control	9.7	C
VS-86	School-Level Academic Reports	9.8	P
VS-87	School’s Own Question Bank ★	9.9	P
VS-88	Offline Mode + Smart Sync ★	10.1	P · P3
VS-89	Data Saver + Audio-Only Mode	10.2	P
VS-90	Language Layers (3 layers)	10.3	P
VS-91	Low-End Device Optimization	10.4	P
VS-92	Academic + Festival Calendar	10.5	P
VS-93	Streaks with Freeze	11.1	P · P4
VS-94	Improvement Badges	11.2	P
VS-95	Daily 5-Minute Challenge	11.3	P
VS-96	Game: MCQ Rush	11.4	P
VS-97	Game: Match-the-Concept	11.4	P
VS-98	Game: Rapid Revision	11.4	P
VS-99	Game: Streak Challenge	11.4	P
VS-100	Game: Boss Level	11.4	P

The spec header describes the catalogue as “101 features, VS-01 to VS-101” — the numbering defines VS-01..VS-100 and counts the core loop as the unifying 101st element. No feature has been invented to fill the gap; see Section 17.3 for Blueprint modules beyond this list.

17.2 Role × Feature Matrix

Primary ownership per feature group (X = primary, x = secondary; (c) = class teacher; (t) = teacher view of student data; (p) = parent mode; (r) = recipient).

Feature group	Student	Teacher	Parent	Admin
Entry VS-01, VS-08			x	X
Entry VS-02, VS-83–85				X
Entry VS-03, VS-82		X		x
Entry VS-04–VS-06	X			x
Entry VS-07 (all roles)	x	x	x	x
Workspace VS-09, VS-14	X	X		
Workspace VS-10	X	X (c)		
Workspace VS-11–13	X	X		
Student VS-15–27	X	x (t)		
AI VS-28–31, 33–36	X			
AI VS-32 (governance)	x	x		x
AI VS-37, VS-38		X	x	
AI VS-39–42	X	x (t)		
AI VS-43–46	X	X (c)		
AI VS-47–50	X	x		
AI VS-51–54	X			
AI VS-55–57		X	x	
AI VS-58, VS-59	X	X	X (p)	
Teacher VS-60–66		X		
Teacher VS-67, VS-68		X		
Teacher VS-69		X	X (r)	
Teacher VS-70, VS-71		X		
Teacher VS-72		X (c)		
Parent VS-73–78			X	
Admin VS-79–87				X
India VS-88–92	X	X	X	X
Motivation VS-93–100	X	x		

17.3 Blueprint-Only Feature List (beyond the VS numbering)

The Blueprint (S5) specifies these modules without VS IDs. They are part of the documented product; cite them as BP-§section:

BP-§47 Personalised Explanation Engine (8 representation modes) · **BP-§48** Learning Remix (mystery / counterexample / story formats) · **BP-§49** Student Learning-State Model (per-concept, revisable) · **BP-§50** Concept & Prerequisite Graph (teacher-correctable) · **BP-§53** What-If Machine (deterministic simulation + AI explanation) · **BP-§55** Chapter-to-Comic Studio (4–6 panel illustrated explanations)

BP-§56 Infinite Textbook Adventure (branching curriculum story) · **BP-§57** Concept Court (two characters argue, student judges) · **BP-§58** Reverse Exam Mode (rubric feedback, rewrite, compare) · **BP-§60** Reality Scanner (curriculum concepts via everyday objects) · **BP-§61** AI Debate Between Past and Future Thinking · **BP-§63** Parent Translator (evidence → plain parent language)

BP-§64 Learning Passport (portable evidence of demonstrated skills) · **BP-§32** AI School Knowledge Compiler (12 outputs from one source) · **BP-§36** Classroom Concept Radar (misconception frequency view) · **BP-§38** Targeted Reteaching Assistant · **BP-§39** Teacher Workload Assistant (7 drafting functions)

BP-§21 Time-Controlled Lesson Release (timezone-correct) · **BP-§67** Hints & Scaffolded Help (5-step ladder) · **BP-§69** Reflection prompts (4 questions, post-activity)

17.4 Open Decisions Register (all sources, consolidated)

Do NOT guess on these; leave open until the team agrees. Sources: Team Guide §14 [S1], Blueprint Part 18 [S5], Vision open questions [S2], Master Spec inline notes [S4].

#	Decision	Options / note	Src
1	Product name (idea matters more than the name)	VidhyaSaathy / ClassOS / other	S1
2	Student login & verification method	Activation code / parent phone / admin approval (never roll number alone)	S1,S2,S4,S5
3	Who adds students	Admin upload / class teacher / self-join	S1,S4
4	Minimum evidence before a prerequisite gap is suggested	Question count + types undecided	S1,S2
5	Band thresholds vs per-concept tracking	Per-concept probably better; not validated	S1
6	Class insights show names or counts only?	Counts-only is the working default	S1,S2,S4
7	How parents receive updates	Daily / weekly / on-event	S1
8	How a session becomes complete	Slides viewed + quiz (PROPOSED: check mandatory)	S1
9	Target school for the first pilot	Own / partner / other	S1
10	Final logo artwork, font, icon set	6-part logo spec §2.4	S5,S3
11	Final tagline (choose ONE)	“Learn Grow Together” / “Grow at your own pace.”	S3,S5

12	Initial curriculum	First class, subjects, chapters	S5
13	AI provider, deployment, budget	Not selected	S5
14	Who may approve AI-generated lessons/explanations	Approval rules undecided	S5
15	Exact parent-visible fields	To finalise	S5
16	When AI feedback is allowed vs mandatory teacher review	Assessment rules undecided	S5
17	Languages in the first release	English/Hindi/Hinglish+?	S5
18	Which events trigger notifications	To define	S5
19	Data retention / deletion / export rules	To define	S5
20	Offline capability in the first pilot	Needed or not	S5,S4
21	First "signature AI feature" after the core loop works	To select	S5
22	Success metrics for usefulness evidence	To define	S5
23	Dashboard shows all subjects or nearest deadline only	OPEN	S5
24	School-level teacher tooling vs single teacher account	Assumption to validate	S2
25	Whether schools allow classroom recordings	Assumption to validate	S2
26	Whether students use after-school sessions regularly	Assumption to validate	S2
27	Whether fragmentation/coaching dependence is as described	Hypothesis — needs research	S2

17.5 Cross-Reference Map of Key Workflows

The core loop

VS-60/61 → VS-62/63 → VS-15/16 → VS-17/18 → VS-27/39-42 → VS-28-31/58 → VS-60/61 (repeats; see Section 1.2)

Recovery chain (absent student)

trigger → VS-20 → VS-12 (segment) → VS-11 (notes) → VS-16 (mini session) → VS-18 (5-question check) → pass = class pace / fail = VS-30 (root cause) → VS-31 (plan)

Doubt chain (night questions → morning insight)

VS-35/VS-36 (voice/photo) → VS-29/VS-28 (Socratic/context answer) → VS-37 (anonymous clustering) → optional VS-38 (named teacher inbox) → VS-60/VS-61 (morning dashboard)

Publishing chain (one effort → reusable asset)

VS-67 (AI skeleton) → VS-63 (builder) → VS-62/64/65 (notes, recording, homework) → VS-66 (grading) → VS-68 (next-year reuse) → VS-87 (question bank grows)

Parent chain (evidence → support)

VS-66/VS-18 results → VS-69 (one-tap update) / BP-§63 (translator) → VS-73 (dashboard) → VS-74 (digest) / VS-75 (ask-this prompt) / VS-76 (WhatsApp) → VS-59/VS-77 (PTM rehearsal)

Personalisation chain (evidence → next step)

VS-18 → VS-39 (confidence) + VS-40 (DNA) + VS-42 (avoidance) → VS-30 (gap) → VS-31 (plan) → VS-23 (difficulty) → VS-44 (remix) / VS-45 (enemy questions) → VS-100 (boss unlock)

PSYCHOLOGY-SAFETY CHAIN — RULES THAT GUARD EVERY AI FEATURE

VS-32 (honesty) + VS-33 (frustration-aware, no biometrics) + VS-07 (RBAC) + VS-78 (parent boundaries) + Part 13 rules.

PART 18 · CLOSING

Glossary, Team Workspace and Closing

The product's vocabulary, the working process around it, and the final definition.

18.1 Glossary

Activation Code — one-time school-issued code that lets a verified student unlock their account (VS-04).
Avoidance Pattern — weeks of skipping one concept; the fear signal tracked by VS-42.
Band — internal 0-25 / 25-50 / 50-75 / 75-100 routing tier (VS-22); never shown as a label.
Boss Level — game mixing all chapter concepts; pass unlocks the next chapter (VS-100).
Concept — a single idea within a chapter, e.g. acceleration.
Concept Heat-Map — class grid coloured by % of students correct (VS-61).
Confidence Meter — pre-question "Sure / I think / Guess" capture paired with outcomes ("Pakka Meter", VS-39).
Digital Twin — a simulated next test composed from current mastery (VS-54).
Distance vs Displacement — the canonical Motion-chapter example used across the sources.

Emphasis Moment — a point in a recording where the teacher stressed, repeated or warned (VS-55).
Enemy Question — a previously-missed type that returns for defeat — 3 correct in different forms ("Dushman Sawaal", VS-45).
Guided Session — teacher-created, time-controlled study unit with the fixed 10-step shape (VS-16).
Hinglish Bridge — English material explained in Hinglish, technical terms kept in English (VS-47).
Learning Passport — portable record of demonstrated skills with evidence (BP-§64).
Learning Recovery Path — structured catch-up sequence after a missed lesson (VS-20 / BP-§52).
Learning-State Model — per-concept, revisable estimate of understanding (BP-§49).
Mastery Estimate — a per-concept estimate based on evidence, always open to revision.
Mistake DNA — classification of errors into careless / concept / speed / reading ("Galti DNA", VS-40).

Moat Test — "Can ChatGPT already do this?" — the gate every AI feature must fail to be worth building.
Prerequisite — an earlier concept needed to understand a later one.
Question Bank — the school's own, ever-growing approved question asset (VS-87).
Rote Detection — cross-form questioning exposing definition-vs-application mismatch ("Ratta Detector", VS-41).
Socratic Mode — the AI guides the student to the answer themselves (VS-29).
Streak Freeze — automatic streak preservation during illness, exams, holidays (VS-93).
Syllabus GPS — class-position vs self-position vs exam-date triangulation with re-routing (VS-58).
Teacher DNA — the data profile of a teacher's emphasis and question patterns, teacher-approved (VS-57).
Viva — oral examination practice against a voice AI (VS-51).

18.2 Team Workspace Protocol (S1 §15, S2 Part 12)

This section is the working process around the product — none of it is approved feature scope until the team marks it approved.

- **Idea backlog** (one row per idea): Idea ID / feature or idea / what it is / problem it solves / who benefits / why it might help / Priority (High, Medium, Low, Undecided) / Status (New, To Discuss, Approved, Rejected, In Development, Done) / team notes.
- **Feature deep-dive sheet** (copy per idea worth understanding): feature name; date added; proposed by; problem being solved; target user; how it would work; step-by-step user journey; what the user provides or does; what the system does; expected result; connection to existing workflows; dependencies; possible technical requirements; challenges and risks; how to test it; team comments; final decision and decision date.

- **Open questions sheet:** decision required / why it matters / options / current thinking / final decision / owner / status. (Working content consolidated in Section 17.4.)
- **Validation sheet:** assumption / who can help validate / how to test / what was learned / what changes as a result.
- **Roadmap stages:** current concept (this documentation) → approved for first prototype (NONE yet — team to decide) → considered for later versions → long-term possibilities.
- **Action tracker:** task / why it matters / owner / priority / deadline / status / notes. **Next meeting:** date, main topics, decisions made, questions remaining, actions before the next meeting. **Change log:** date / what changed / why / who approved.
- **Standing homework from the Team Guide:** before the next meeting, each member writes ONE thing they would change about the idea and ONE thing they think is missing, and brings both to the call.

18.3 Final Product Definition (S5 Closing)

VidhyaSaathy is an AI-powered, school-connected learning platform that helps students continue learning at home, gives teachers actionable insight into student understanding, and helps parents support progress through clear, evidence-based updates. Its foundation is not the number of AI features. It is the connection between school-approved knowledge, student learning evidence, personalised next steps and teacher intervention.

The cloud companion provides a friendly identity. The guided self-study system provides structure. The AI learning engine provides personalisation. The teacher dashboard provides actionable insight. The parent dashboard provides shared understanding. The long-term ambition: a student does not merely receive an answer, but gets help identifying what they do not understand, learning it in a suitable way, checking whether they improved, and continuing with the right next step.

> *PRODUCT PRINCIPLE: One classroom. Different learning paths. One connected support system.*

18.4 Source Documents & Fidelity Note

- **[S1]** VidhyaSaathy_Team_Guide.pdf — "Your school, online, after the bell rings." Team draft v1, 13 October 2026, Vikrant University Junior Hackathon 2026. Vision, workflows, open decisions, team workspace. (The team mockup inside is titled ClassOS; the name is not final.)
- **[S2]** VidhyaSaathy — Product Vision & Team Workspace.pdf — Oct 13, 2026. Executive summary, product vision, users, Motion example, responsible design, glossary, team workspace templates.
- **[S3]** VidhyaSaathy brand & theme board (image) — cloud mascot reading a book; palette #7FB3FF / #FFC8A3 / #FFF9F5 / #1E3A8A / #E6F0FF; six mascot expressions; "Small steps make big progress"; app UI mock.
- **[S4]** VidhyaSaathy Master Working Spec — features VS-01 to VS-100 across 12 parts, the minute-by-minute day, deliberate cuts, build order.
- **[S5]** VidhyaSaathy Complete Product Blueprint / PRD — Parts 1-18: foundation, brand, entry, dashboards, AI engine, activities, communication, learning graph, end-to-end example, data & security, screens checklists, architecture, build priorities, acceptance criteria, decisions to finalise.



One classroom. Different learning paths. One connected support system.

“Why should the school’s support end when the bell rings?”
The school speaks again — after the bell rings.

VidhyaSaathy

BREAKTHROUGH EDITION · VERSION 1.0 FINAL · 19 OCTOBER 2026

Fidelity note: every feature, workflow, rule, example and open question in this document traces to the five source documents (Part 18.4). Hindi/Hinglish passages were translated to clear English; product vocabulary from the original language is preserved in parentheses. Where sources disagreed on scope, both positions are recorded rather than resolved.